



SPRINGER PUBLISHING

FAMILY NURSE PRACTITIONER CERTIFICATION **Intensive Review**

Plus Interactive Digital Prep

MARIAT. CODINA LEIK

Fourth Edition

Family Nurse Practitioner Certification Intensive Review

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*What a great time to be a nurse practitioner.
This is our decade, girls (and guys). I dedicate this book to you.
What a privilege it is to be able to make a difference in someone's life.
Wow.*

*Of course, my grateful thanks to
Ed, Mary, and Chrissy.*

Contents

<i>Contributors and Reviewers</i>	<i>ix</i>
<i>Preface</i>	<i>xi</i>
<i>Acknowledgments</i>	<i>xiii</i>

I. CERTIFICATION AND EXAM INFORMATION

- | | |
|-------------------------------------|----|
| 1. Certification Exam Information | 3 |
| 2. Question Dissection and Analysis | 21 |

II. PHARMACOLOGY AND HEALTH SCREENING OVERVIEW

- | | |
|---|----|
| 3. Pharmacology Review | 59 |
| 4. Health Promotion, Screening, and Disease Prevention Review | 95 |

III. SYSTEMS REVIEW

- | | |
|--|-----|
| 5. Head, Eyes, Ears, Nose, and Throat Review | 113 |
| 6. Integumentary System Review | 143 |
| 7. Cardiovascular System Review | 181 |
| 8. Respiratory System Review | 219 |
| 9. Endocrine System Review | 245 |
| 10. Gastrointestinal System Review | 263 |
| 11. Genitourinary and Renal System Review | 283 |
| 12. Nervous System Review | 295 |
| 13. Hematopoietic System Review | 319 |
| 14. Musculoskeletal System Review | 333 |
| 15. Psychiatric–Mental Health Review | 359 |

IV. REPRODUCTIVE REVIEW

- | | |
|---|-----|
| 16. Male Reproductive System Review | 381 |
| 17. Female Reproductive System Review | 393 |
| 18. Female Reproductive System Review: Pregnancy and Childbirth | 421 |
| 19. Sexually Transmitted Diseases and Infections Review | 445 |

V. PEDIATRICS AND ADOLESCENTS REVIEW

20. Pediatrics Review: Newborns and Infants 465

21. Pediatrics Review: Toddlers (Age 2–3 Years) 487

22. Pediatrics Review: School-Age Children (Age 4–11 Years) 495

23. Adolescents Review 507

VI. GERONTOLOGY REVIEW

24. Body and Metabolic Changes in the Older Adult 521

25. Common Disorders in Geriatrics 529

VII. PROFESSIONAL ROLE REVIEW

26. Ethical Guidelines and Advanced Practice Law 551

27. Professional Roles and Reimbursement 565

28. Cultural and Spiritual Health Beliefs 573

29. Nursing Research 581

30. Evidence-Based Medicine and Epidemiology 589

VIII. PRACTICE TESTS

31. Practice Test I 599

32. Practice Test I: Answers 623

33. Practice Test II 643

34. Practice Test II: Answers 667

35. Practice Test III 687

36. Practice Test III: Answers 715

37. Practice Test IV 735

38. Practice Test IV: Answers 763

Abbreviations and Acronyms 785

Resources 793

Index 811

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Preface

Welcome to the fourth edition of *Family Nurse Practitioner Certification Intensive Review*. This review book is designed to help you study effectively and efficiently for the family nurse practitioner certification exams of both the American Academy of Nurse Practitioners Certification Board (AANPCB) and American Nurses Credentialing Center (ANCC).

The goal for all of my nurse practitioner (NP) certification review books is to provide you, the NP test-taker, with a tool to effectively and rapidly review and prepare for your exam. This book is also a valuable resource for current NP students and educators. Many students credit these books with helping them pass not only their certification exam but also quizzes and tests when they were students. Students report that the review book has helped them successfully pass their school's dreaded "exit" exam.

The book's format remains the same as in the popular and best-selling first three editions, with the addition of several new features. The book is styled as a "mega-review" study guide because it combines six different resources in one:

1. Certification exam information that is highly specific for both the AANPCB and ANCC exams
2. Useful test-taking techniques to help you prepare and strategize for exam day
3. Question dissection and analysis tools to help you break down questions for further study and hone critical-thinking skills
4. A thorough review of pharmacology; health promotion, screening, and disease prevention; and primary care diseases and conditions, with pertinent normal physical exam findings, danger signals, exam tips, and clinical pearls
5. **New:** End-of-chapter "knowledge check" questions designed to help you assess knowledge retention and application
6. **New:** Four full practice tests based on the blueprints for the AANPCB and ANCC certification exams, with answers and detailed rationales

The fourth edition has been significantly updated and features new disease topics, tables, procedures, and updated clinical content that relate to the latest exam questions, as well as many full-color images of dermatologic diseases and other conditions. In addition, it includes *hundreds* of new practice questions.

Readers receive 6 months of free digital access to the interactive digital version on ExamPrepConnect (see access details on inside front cover). Features include:

- All of the high-quality content from the book
- Personalized study plan based on exam date
- Ability to study by topic area to identify strengths and weaknesses

- Timed practice tests that simulate the test-taking environment
- Discussion board to connect with the nurse practitioner exam-prep community
- Games to make studying fun

This book and my *Adult-Gerontology Nurse Practitioner Certification Intensive Review* have become the most sought-after and relied-upon review books for use in studying for both the AANPCB and ANCC certification exams. The contents and procedures in this book are designed for review purposes only. They are not intended for use in clinical practice.

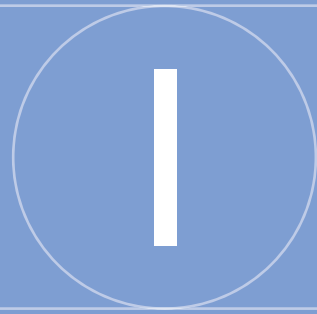
If you have any comments or suggestions for this book or would like me to teach a review course at your institution, please contact me at npreviews@gmail.com. If you would like to attend one of my webinars or live-review courses, visit my website at www.npreview.com, which is frequently updated with exam information, updates, and new treatment guidelines.

Maria T. Codina Leik

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Certification and Exam Information

Certification Exam Information

1

AMERICAN NURSES CREDENTIALING CENTER

www.nursingworld.org/ancc

Credential: FNP-BC

Toll-Free Phone: (800) 284-2378

Mobile App: MYANCC (Apple Store/Google Play)

Electronic Transcripts: aprnvalidation@ana.org

The American Nurses Credentialing Center (ANCC) is the independent credentialing body of the American Nurses Association (ANA). It is the largest nurse credentialing organization in the United States. The current edition of the ANCC Family Nurse Practitioner (FNP) exam was released May 22, 2019.

The ANCC currently offers certification exams for the following nurse practitioner (NP) specialties: Family NP (FNP-BC), Adult-Gerontology Primary Care NP (AGPCNP-BC), Adult-Gerontology Acute Care NP (AGACNP-BC), and Psychiatric-Mental Health NP (PMHNP-BC). You must first establish an online account at the ANCC website before you can apply for any of the exams. You can use the mobile app MYANCC to apply or renew your certification and keep track of your continuing-education (CE) hours.

Prometric Computer Testing Centers

www.prometric.com/ANCC

Toll-Free Phone: (800) 350-7076

Special Conditions Department (Special Accommodations): (800) 967-1139

The ANCC uses Prometric Testing Centers to administer its exams. Before you can schedule an appointment, you must have your candidate eligibility number; this is provided in the Authorization to Test Notice letter sent to qualified candidates who meet eligibility requirements and have completed their application documentation.

You can schedule, reschedule, cancel, and/or confirm an appointment at the website portal or by telephone. Prometric has 14,000 test centers in 180 countries, including the United States, Puerto Rico, the U.S. Virgin Islands, Canada, Venezuela, and the United Kingdom. There are also test centers in Africa, Asia, and Europe.

International Testing for Military Personnel

Military nurses who want to take the exam outside the United States are allowed to take their exams at one of Prometric's "global" testing centers.

Qualifications

One must graduate from an approved master's, postgraduate, or doctoral degree program with a specialization in family NP (plus the required minimum of 500 faculty-supervised clinical practice hours) from an institution of higher learning. The program must be accredited by the Commission on Collegiate Nursing Education (CCNE) or the Accreditation Commission for Education in Nursing (formerly known as the National League for Nursing Accrediting Commission). In addition, the candidate must possess an active RN license in any state or territory of the United States. The ANCC will, on an individual basis,

accept the professional legally recognized equivalent (of NP status) from another country if the applicant meets its certification criteria. If the applicant is educated and/or licensed outside the United States, there are additional requirements.

AMERICAN ACADEMY OF NURSE PRACTITIONERS CERTIFICATION BOARD

www.aanpcert.org

Credential: FNP-C

Toll-Free Phone: (855) 822-6727

Certification Administration: (512) 637-0500

Email: certification@aanpcert.org

Electronic Transcripts: transcripts@aanpcert.org (via secure electronic transmission from registrar's office)

The American Academy of Nurse Practitioners Certification Board (AANPCB) is an independent organization that is affiliated with the AANP and the American Academy of Emergency Nurse Practitioners (AAENP).

The AANPCB is currently offering three specialty exams: Family NP (FNP-C), Adult-Gerontology Primary Care NP (AGNP-C), and Emergency NP (ENP-C). The current FNP, AGNP, and ENP exams were released in February 2018. Their Adult NP exam was retired in December 2015, and the Gerontologic NP exam was retired in December 2012.

You must first establish an online account at their website before you can apply for the exam. Applicants enrolled in an MSN or post-master's certificate program may begin the application process 6 months before completion of the program. If you are enrolled in a DNP program, you can begin the application process as early as 1 year prior to completion of the program. But you must complete all the NP program didactic courses and supervised clinical practice hour requirements before you can sit for the exam. In addition, an official transcript showing the doctor of nursing practice (DNP) degree awarded and conferral date is required to release your exam score. The certification start date will be the date the score is released (not the date the examination was taken).

PSI Computer Testing Centers

www.psiexams.com

Toll-Free Phone: (800) 733-9267

Fax: (702) 932-2666

The AANPCB's exams are administered by PSI Services LLC, which has more than 300 locations across the United States and Canada. Locations are listed on the PSI scheduling portal. Approved, qualified candidates who have completed their applications (final transcripts included) are sent an authorization to test by email from the AANPCB. Approximately 1 day after you receive the AANPCB email, PSI Services will send you your PSI eligibility ID number from support@psionline.com, providing a link to their online scheduling system (to schedule/reschedule/cancel/confirm). You can also register by phone (800-733-9276). Important information about applying for the AANC and AANPCB exams is provided in Table 1.1.

EXAM BACKGROUND INFORMATION

The ANCC usually releases a new edition of their exam every 2 years. Each ANCC exam edition has several versions. The AANPCB is now releasing a new edition of their exam annually in January; each edition has at least two versions. If you have failed the exam and retake it, you will get a different version of the exam. The version of the exam that you are taking is assigned by the testing-center computer system. For example, you may be taking a different version of the exam than your classmate even though both of you are taking the same edition of the exam on the same date and time at the same testing center.

Table 1.1 Application Timeline for Certification Exams

Timeline	Recommended Activities Before Graduation
6 months before graduation	Open an account on the ANCC and/or the AANPCB website and start the application process. Find out about the certification requirements. Download and read the General Testing Booklet, Test Content Outline, and Reference List. Visit your SBON website and read the licensure requirements for NPs in your state. Download the licensure application.
4–6 weeks before graduation	Order three to four copies of your official final transcript.
2–3 weeks before graduation	Ask the NP director of your program to sign the ANCC Education Validation form.

AANPCB, American Academy of Nurse Practitioners Certification Board; ANCC, American Nurses Credentialing Center; NP, nurse practitioner; SBON, state board of nursing.

Exam Format

The formats on the ANCC exam include “drag and drop”; multiple-choice question with five or six answer options (you are asked to select all answers that apply); “hot spot” (a picture is presented that is divided into four quadrants, and you click on the correct quadrant as the answer); and colored photographs of skin and eye conditions; EKG strips; and chest x-ray films. These new formats are discussed in Chapter 2. Most questions on the ANCC exam are still multiple choice with four answer options.

Similarly, most AANPCB exam questions are the classic multiple-choice format with four answer options. There may be one EKG strip, a chest x-ray film, and some multiple-choice questions with five to six answer options (you are asked to select two to three answers). The exams are computer-based tests (CBTs). Unlike the NCLEX-RN® exam, which is a computer-adaptive test, the NP exams do not shut down automatically after you have earned enough points.

Test Content Outline and Reference List

Both the ANCC and the AANPCB provide test content outlines (TCOs) and a reference list on their website. It is important that you read these documents carefully and concentrate your studies on the designated topic areas.

Total Number of Questions and Time

- *ANCC FNP Exam (released May 2019)*: There are 175 total questions. Only 150 questions are graded; 25 are pretest questions that are not scored.
 - **Total Time**: Total counted testing time is 4 hours.
- *AANPCB: FNP Exam (released January 2020)*: There are 150 total questions. Only 135 questions are graded; 15 questions are pretest questions that are not scored.
 - **Total Time**: Total counted testing time is 3 hours.

Computer Tutorial

For both exams, each testing session starts with a computer tutorial session of 10 to 15 minutes. This time is not counted as part of the testing time; it is “free” time. When the tutorial time expires, the computer will automatically start the exam, and the first question will appear on the screen. The countdown clock (upper corner of the screen) will start counting down the time. The countdown clock will not stop for “breaks,” but if you need a break, inform a testing center staff member who is monitoring the testing area. Watch the time clock closely because when the allotted testing time expires, the computer will automatically shut down the exam.

Time Allotted per Question

For both the ANCC and AANPCB certification exams, each question is allotted approximately 60 seconds. If you find yourself spending too much time on one question, pick an answer (best guess), then “mark” it so that you can return to it later. Never leave any question unanswered. If you guess correctly, it will be counted; if you leave it blank, it is marked as an error (0 points).

Sample Exam Questions

I highly recommend that you try the free sample exams at the ANCC’s website even if you are planning to take only the AANPCB exam. The ANCC’s sample exam is web-based and contains 25 questions. It can be taken as many times as desired at no charge (and it is scored every time). The AANPCB also has free questions, which are listed on the last two pages of their candidate handbook (no answer key is provided). If you want to familiarize yourself with the format of the AANPCB exam, you can purchase their 75-item practice examination (timed 90 minutes), which is taken online and graded. It can be taken only once. The ANCC also sells sample exam questions, but their free web-based sample exam provides a good opportunity to practice using their exam formats.

COMPARING THE ANCC AND AANPCB EXAMS

1. What are the passing rates for the two exams?

ANCC FNP exam: 87%

AANPCB FNP exam: 86%

2. What is the major difference between the two exams?

There are two major differences between the two exams. First, the ANCC exam contains nonclinical questions. Second, question formats, such as drag and drop, photographs, EKG strip, chest x-ray film, and multiple-choice questions with five to six answer options, were added beginning with the 2016 ANCC exams.

The questions on the AANPCB exam are all on clinical topics. The current version (January 2020) does not contain nonclinical topics. Most of the questions are the classic multiple-choice format with four answer options, but there may be some multiple-choice questions that have five to six answer options (you are asked to select two to three answers). An EKG strip and/or chest x-ray film is not always included on the exam (AANCB and ANCC).

3. Are there apps from the Apple Store and Google Play?

AANC: The ANCC has an app that can be used to apply for the exam.

AANPCB: The AANPCB does not have an app currently, but it may have one in the near future.

4. What are the new commands in the ANCC Prometric exams?

You can now “highlight” words (keywords) to help you concentrate on the best clues (that you picked). You can also use the “strikeout” command, which allows you to cross out (eliminate) the answer option(s) that you think is(are) wrong (distractors). These two new commands are not yet available for the PSI AANPCB exam.

5. If I join their respective membership organizations, will I receive a discount on my fees?

ANCC: To be eligible, you need to be a member of the ANA. If you belong to a state nursing association, you may already be a member of the ANA. Check with your state’s nursing association. To apply for membership, apply online at the ANA’s website (www.nursingworld.org). *A word of warning:* If you are not a member, you have only 5 business days after you apply for the exam to claim the membership discount. Exam discount claims received after this time are not allowed, and “refunds will not be issued.”

AANPCB: To be eligible, you must be a member of the AANP (www.aanp.org). If you already applied for the exam as a nonmember, you have up to 30 days to apply for membership. To receive the membership discount (as a refund), inform them by email that you are a member (certification@aanpcert.org).

6. How do I apply for the ANCC or AANPCB certification exam?

For both agencies, you must initially open an online account on their website. To avoid loss of information, it is best to complete your online application within 30 days of starting it. The application

forms are in editable PDF format so you can enter the information directly on your computer and then print a hardcopy for your records. If you apply by mail, do not forget to photocopy the entire contents of your application package. If you plan on using the U.S. Postal Service to mail your package, it is recommended that you pay the extra fee for registered mail.

ANCC: The ANCC accepts applications online and by email, standard mail, fax, and their mobile app MYANCC (Apple Store/Google Play).

AANPCB: The AANPCB accepts applications online and by mail. Applying online does not expedite processing. The organization automatically charges a nonrefundable paper application processing fee for all paper applications (email, mail, and fax). There is no charge for processing of supporting documents for certification and recertification (e.g., unofficial transcripts, RN license).

7. How early can I apply for the certification exam?

ANCC: The ANCC allows students to apply as early as 6 months before graduation. It will not process an application until payment has cleared. A completed application must contain a copy of your RN license, an official copy of your final transcripts, and the original copy of the Validation of Education form, signed by your NP program director. A Gap Analysis form (on website) is needed for all postgraduate certificates or doctoral degrees and should be sent with the Validation of Education form.

AANPCB: Applicants enrolled in an MSN or post-master's certificate program may apply 6 months prior to graduation. DNP students may apply up to 12 months before graduation, but the AANPCB will not release the exam scores or grant certification until the MSN or DNP degrees are issued. If you apply before you graduate, you are required to submit an official copy of your current transcript (the interim transcript). The interim transcript is your most recent transcript, which contains all the coursework that you have completed up to that date. When the official final transcript is released, either you or the registrar's office of your school can mail or email it directly to the AANPCB. In addition, the AANPCB requires a hard copy of your RN license. Do not forget to download the State Board of Nursing Notification form from the board's website so that it can release your exam scores to your state board of nursing (SBON).

8. How long does it take to process a completed application?

ANCC: It takes about 4 to 6 weeks to process an application. Be warned that the ANCC will not process an application until it is complete and all required documents have been received (copy of RN license, original of the signed education validation form, final transcript). If there are problems with your application, an email will be sent to you. Answer all the questions on the forms, and do not leave any blanks.

AANPCB: It generally takes between 4 and 6 weeks after you have paid the fees and submitted all the required documentation to process an application. Fill out all the questions, and do not leave any blanks or forget signatures. An error can delay your application significantly.

9. What is the ANCC's "expedited application" process?

This option shortens the processing time to 5 business days (\$200 nonrefundable fee). Download and print the Certification Expedite Review Request form from the ANCC website and fax it to (301) 628-5233. If your school has not released your final transcript, it will delay the expedited application. The Validation of Education form can be signed by electronic signature (give your NP director's email address), and an electronic copy of your final transcript can be emailed directly by secure transmission from your school's registrar office (aprnvalidation@ana.org). Call to verify that the ANCC has received your faxed application.

Note: Mailing the ANCC "expedited" form will delay processing; it should be faxed. If you both fax and mail it, you will be charged the processing fee twice.

The AANPCB does not expedite applications.

10. What are the final transcript and the "official" transcript?

The final transcript is the one that is issued after you graduate from your NP program. It should indicate the type of degree you earned along with the NP specialty (e.g., family or adult-gerontology NP).

A transcript is considered official only if it remains inside the sealed envelope in which it was mailed from your college registrar's office or if it is mailed directly from their office to the certifying agency. An

electronic copy of your transcript can be transmitted directly by secure transmission from the registrars' office (aprnvalidation@ana.org). Order at least three to four copies of your final transcripts and keep the extras unopened to keep them official. Open one copy (yours) and check it for accuracy.

11. What is the ANCC Validation of Education form?

This one-page form is a requirement for ANCC applicants only. Most NP programs have this document signed in bulk by the current director of the NP program. Ideally, it should be distributed to the students before the final day of school. You must send the original signed form (not a copy) to the ANCC with your application.

Note: It is probably a good idea for all NP students to get this document signed, even if they plan to take only the AANPCB exam. Because the ANCC requires this form, it will save you precious time in the future if you decide to apply for their exam. You always have the option of applying quickly for the "other" certification exam if you fail the first. Do not forget that you will need another official final transcript (another good reason why you should have at least four copies).

12. Is it possible to take the exams at any time of the year?

You can take both certification exams at any time of the year, whenever the Prometric and the PSI testing centers are open.

13. When should I schedule my exam?

Do not wait until the last minute to schedule your exam because the testing centers in your area may no longer have the date (or the time) that you desire. Morning time slots tend to get filled very quickly. Both Prometric and PSI testing centers allow test takers to schedule, reschedule, or cancel an appointment on their website or by phone, but you should give them notice.

Prometric (ANCC): The processing fee is charged if <30-days' notice is given.

PSI (AANPCB): Requires a minimum of at least 2 days in advance to reschedule or cancel an exam (before 5 p.m. Pacific Standard Time).

Note: Avoid scheduling yourself at the time of day when you tend to get tired or sleepy. For most, this is usually after lunchtime. Simply picking the wrong time of day can cause you to fail the exam, sometimes by as little as 2 points.

14. What should I do if the time slot or date that I want is no longer available?

Look for another testing center as soon as possible. For some, it may mean a long drive to another city, but it may well be worth the extra time and effort if your authorization-to-test letter is about to expire.

15. Can I take the exam if my name does not match my primary picture identification?

The name that you used to register must match exactly the name on your primary picture identification with signature. If it doesn't, you will not be allowed to test. If you need to modify your name, call the ANCC as soon as possible prior to scheduling your appointment.

16. If I have a disability, how can I obtain special testing accommodations?

You can download the Special Accommodations form from each organization's website (ANCC and AANPCB). It requires a description of the disability and limitations related to testing. It should be dated and signed by the provider. For example, if a student has severe test-taking anxiety, a special accommodation (if approved) is an extension of the testing time. Check the agency's website for further instructions.

COMMON QUESTIONS ABOUT ANCC AND AANPCB EXAMS

1. What are the passing scores?

ANCC: The passing score is 350 points or higher. ANCC scores range from 100 to 500 points.

AANPCB: The passing score is 500 points or higher. AANPCB scores range from 200 to 800 points.

2. Will I find out immediately if I pass (or fail) the exam?

Yes, you will. After you exit the testing area, ask the proctor to print a copy of your results (pass or fail format). This is the “unofficial” score. If you pass the exam, you will not get a number score. According to the AANPCB, a preliminary “pass” score at the testing center is not considered official notification. It does not indicate active certification status, and it may not be used for employment or licensure as an NP. A letter with your official score will be mailed to you at a later date, indicating the type of certification and the starting date. If you fail the exam, you will receive a scaled number score with diagnostic feedback information for each domain such as:

- *Low:* Your score from this content area is below the acceptable level.
- *Medium:* Your score from this content area is marginally acceptable.
- *High:* Your score from this content area is above average.
- Concentrate your studies on the domain/category(s) in which you score “low” or “medium.”

3. How many times a year can I retake the FNP certification exam?

ANCC: You can take this exam up to three times in a 12-month calendar period. You must wait at least 60 days between each exam. If you fail, you can reapply online (retest application), but wait about 5 days (after taking your exam) before you reapply online.

AANPCB: You can take the exam up to two times per calendar year. Although there is no wait-time requirement prior to retesting, the AANPCB requires test-takers to take 15 contact hours in their area(s) with the weakest score (your “areas of weakness”). Reapply for the exam online by using the Retake Application form.

Note: The CE hours that you took before sitting for the certification exam are not eligible. The contact hours must be taken after you failed the exam. If you took a review course, it can be retaken again with the new dates. When you have completed the required 15 contact hours, email or fax the certificate of completion as proof. Call the AANPCB certification office first before faxing any documents.

4. What happens if I fail the ANCC or AANPCB exam the second time?

You must resubmit a full application along with all the required documentation (like the first time) and pay the full test fee.

5. What happens if my Authorization-to-Test letter expires, and I have not taken the exam?

Immediately call the certifying agency for further instructions. Both the ANCC and the AANPCB will allow you to reschedule your exam (one time only). You must pay a rescheduling fee. If you fail to test on your extended testing window, you must reapply like a new applicant and pay the full registration fee again.

6. Is it possible to extend the testing window?

ANCC: You can request (one time only) a new 90-day testing window. It must occur <6 months from the last day of the initial testing window. Your request for extension to the ANCC should be received before the initial 90-day testing period expires.

AANPCB: You can request an extension (one time only) by sending an email to certification@aanpcert.org. If an extension is granted, it will be for 60 days only. If you do not retake the exam during that time, you must reapply to take the examination (like the first time) and pay the full fee.

7. Is there a penalty for guessing on the exam?

No, there is no penalty for guessing. If you are running out of time, answer the remaining questions at random. Do not leave any questions “blank” or without an answer, because this will be marked as an error (0 points). You may earn a few extra points if you guess correctly, which can make the difference between passing and failing the exam.

8. Is it possible to return to a question later and change its answer?

Yes, it is, but only if you “flag” the question. You will learn about this simple command during the computer tutorial at the beginning of the exam. “Flagging” a question allows the test taker to return to the question at a later time (if you want to change or review your answer). On the other hand, if you indicate to the computer that your answer is “final,” then you will not be allowed to change the answer.

Do not worry about forgetting to “unflag” the questions if you run out of time. As long as a question has an answer, it will be graded by the computer.

9. Does the certifying agency inform the SBON of my certification status?

The AANPCB and ANCC will not automatically send you scores to the SBON. For both certifying agencies, you must sign their Release of Scores consent form when you apply for the exam.

10. How can I obtain an “official” verification of my national certification for my employer?

Both agencies have verification request forms on their websites. If you just took the exam, the AANPCB recommends you wait at least 10 days after your testing date. A nominal fee is charged.

11. I am getting married (or divorced). Is there a problem with changing my legal name after I applied for the exams using my maiden or married name?

Yes, there may be a problem. The name that you use in your application must be the same as the one listed on your primary ID (e.g., your unexpired driver’s license, passport, or military ID). Check the agency’s website or call for more details. If you get married after you applied for the exam, it is better to wait until you have taken and passed the exam before changing the name on your driver’s license.

COMMON QUESTIONS ABOUT CERTIFICATION RENEWAL

1. How long is my certification valid?

Certification from the ANCC as well as the AANPCB is valid for 5 years. A few months before your certification expires, both the ANCC and the AANPCB will send you a reminder letter. “Failure of the certificant to receive (his or her) renewal notice does not relieve the certificant of his or her professional responsibility for renewing their certification.”

2. Can I still use my professional designations after my certification expires because I forgot to renew it?

No, credentials cannot be used once they have expired.

3. What should I do if I change my legal name, have a new email address, have a new telephone contact number, or move to a new address?

You need to update your online account at the ANCC or the AANPCB website within 30 days if there are any changes in your contact information, such as your legal name, residence address, email, or phone number.

4. Is there a time period that must be followed to renew certification?

Yes, there is. All of the contact hours and practice hours must be completed within the current 5-year certification period.

5. When should I submit my renewal application?

The process can be started as early as 12 months before your certification expires. It is best to start early because if you do not have enough CE/clinical/pharmacology contact hours, you still have enough time to obtain them. Recertification applications should be submitted no later than 12 weeks prior to the expiration date of the current certification to allow time for reviewing, processing, and issuing the new certificate before the expiration of the current certification.

Note: Save a digital or hard copy (or other proof) of all your CE hours and clinical practice hours. If you are audited, you will be asked to submit proof of your CE hours (i.e., certificates) and clinical practice hours.

6. How many clinical hours of practice are required to renew my certification every 5 years?

The ANCC no longer requires clinical practice hours for recertification, although the AANPCB does require 1,000 hours of clinical practice in your area of specialty (completed in the previous 5 years before

recertification), plus 100 hours CE, of which 25 hours must be pharmacology CE. If you plan to use your clinical volunteer hours as an NP, keep a notebook to record each clinical site's address, the number of hours that you practiced, and the name/signature of the supervising NP/physician. If you do not have enough clinical hours, another method you can use to recertify is to retake the examination combined with the required CE hours. For adult nurse practitioners (ANPs) and gerontological nurse practitioners (GNPs), you do not have the option to recertify by retaking the examination, because the exams for your specialty have been retired.

7. How many CE contact hours are required for recertification?

ANCC: 75 contact hours of CE are required. You also need one or more of the eight ANCC renewal categories (i.e., academic credits, presentations, preceptor hours, quality improvement project, professional service, practice hours, retake exam, publication or research).

AANPCB: 100 contact hours of CE are required.

8. How many pharmacotherapeutic contact hours are required per renewal cycle?

ANCC: Of the 75 contact hours, 25 of those hours must be in pharmacotherapeutics. If you double the 75 (150 contact hours), only 25 contact hours are required.

AANPCB: Of the 100 contact hours, 25 contact hours of advanced pharmacology are required.

9. What happens if my certification lapsed (I forgot to renew it)?

There is no "grace period" or backdating. On the day when your certification expires, you are prohibited from using your designation after your name. In addition, you will have a gap in your certification dates. The ANCC recommends that you check with your state licensing board to determine whether you can continue to practice as an NP. They also recommend that you check with your employer and with the agencies that are reimbursing your services.

10. If my specialty certification exam was retired, can I continue with my specialty designation?

Yes, you may. You can renew your certification only through CE and clinical hours. If you let your certification lapse, you no longer have the examination option. Unless there are other options in the future (check the websites for updates), only students who have graduated from an AGNP program are permitted to take that exam. ANPs and GNPs are not allowed to take the AGNP exam.

11. Is there any reciprocity between the ANCC and the AANPCB?

No, there is not. Both the ANCC and the AANPCB discontinued their reciprocity program many years ago.

EXAM QUESTION CLASSIFICATION

American Nurses Credentialing Center

See Table 1.2 for the ANCC FNP exam domains.

Table 1.2 ANCC FNP Exam Domains

FNP Exam (Released May 2019)	% of Questions	Number of Questions
Assessment	21	31
Diagnosis	26	39
Clinical Management	43	65
Professional Role	10	15
Total	100	150 scored questions

ANCC, American Nurses Credentialing Center; FNP, family nurse practitioner.

American Academy of Nurse Practitioners Certification Board

See Table 1.3 for the AANPCB FNP exam domains and Table 1.4 for breakdown by patient age group.

Table 1.3 AANPCB FNP Exam: Domains

FNP Exam (Released February 2018)	% of Questions	Number of Questions
Assessment	36	48
Diagnosis	24	33
Plan	23	31
Evaluation	17	23
Total	100	135 scored questions

ANCC, American Academy of Nurse Practitioners Certification Board; FNP, family nurse practitioner.

Table 1.4 AANPCB FNP Exam: Age Groups

Age Group	% of Questions	Number of Questions
Prenatal	3	4
Pediatrics (includes newborn and infant)	14	19
Adolescent (early/late)	18	24
Adult	37	50
Geriatric	21	29
Elderly	7	9
Total	100	135 scored questions

AANPCB, American Academy of Nurse Practitioners Certification Board; FNP, family nurse practitioner.

FAST FACTS: ANCC AND AABPCB EXAMS

1. The ANCC FNP exam has approximately 10% nonclinical questions, or about 15 questions (out of 150 graded questions).
2. The current AANPCB FNP exam released in January 2020 does not contain nonclinical topics.
3. The ANCC and AANPCB FNP certification exams are designed for entry-level (not expert-level) practice. Most test-takers who sit for the FNP certification exams are new graduates.
4. There are 25 pilot-test questions on the ANCC exam and 15 pilot-test questions on the AANPCB exam. These questions are not graded. There is no way to identify the graded questions from the pilot-test questions.
5. New clinical information (treatment guidelines or new drugs) released within the 10 months prior to the current exam will most likely not be included.
6. Keep in mind that the questions will be on primary care disorders (e.g., primary care clinics, public health clinics). If you are guessing, avoid picking an exotic diagnosis as an answer.
7. The AANPCB exams will list the normal lab results when they are pertinent to a question.
8. Unlike the AANPCB, the ANCC does not list the normal lab results. If you are taking the ANCC exam, it is important to memorize some normal lab results and to write them down on scratch paper during the computer tutorial time period.
9. Expect a few questions regarding pregnancy-related topics (obstetrics).
 - *Example:* Uterine involution is when the uterus contracts and shrinks in size until it returns to its prepregnancy state; the process takes about 6 weeks.
10. Learn the significance of abnormal lab results and the type of follow-up needed to further evaluate the patient.
 - *Example:* An elderly male patient complains of a new-onset, left-sided temporal headache accompanied by scalp tenderness and indurated temporal artery. The NP suspects temporal arteritis. The screening test is the sedimentation rate, which is expected to be much higher than normal (elevated value).

- *Example:* A patient with an elevated white blood cell (WBC >11,000/mm³) count accompanied by neutrophilia (>70%) and the presence of bands (“shift to the left”) most likely has a serious bacterial infection.
- 11. Expect to find many culture-related questions on the ANCC exam.
 - *Example:* If a traditional Muslim woman refuses to undress and use a hospital gown, an alternative is to examine her with her clothes on (modified physical exam).
- 12. The other cultures that may be addressed on the ANCC exam are Mexicans, Cubans, Chinese, Cambodians/Filipinos, and Native Americans (e.g., Navajo). Cultural topics are covered in this book in the nonclinical chapters.
 - *Example:* A Navajo woman has an appointment for a follow-up visit, but she does not show up for the appointment. The most common cause for Navajo patients to skip health visits and follow-up visits is the lack of transportation.
- 13. Sometimes there will be one unexpected question relating to a dental injury.
 - *Example:* A completely avulsed permanent tooth should be reimplanted as soon as possible. It can be transported to the dentist in cold milk (not frozen milk).
- 14. There may be a question on epidemiologic terms.
 - *Example:* *Sensitivity* is defined as the ability of a test to detect a person who has the disease. *Specificity* is defined as the ability of a test to detect a person who is healthy (or to detect the person without the disease).
- 15. Learn the definition of some of the research study designs.
 - *Example:* A cohort study follows a group of people who share some common characteristics to observe the development of disease over time (e.g., the Framingham Nurses’ Health Study).
- 16. Several emergent conditions that may present in the primary care area will be on the exam.
 - *Examples:* Navicular fracture, acute myocardial infarction (MI), cauda equina syndrome, anaphylaxis, angioedema, meningococcal meningitis.
- 17. Become familiar with the names of some anatomic areas.
 - *Example:* Trauma to Kiesselbach’s plexus will result in an anterior nosebleed.
- 18. Some questions may ask about the “gold-standard test” or the diagnostic test for a condition.
 - *Example:* The diagnostic or gold-standard test for sickle cell anemia, glucose-6-phosphate dehydrogenase (G6PD) anemia, and alpha or beta thalassemia is the hemoglobin electrophoresis.
- 19. Distinguish between first- and second-line antibiotics.
 - *Example:* A 7-year-old child with acute otitis media (AOM) who is treated with amoxicillin returns in 48 hours without improvement (complains of ear pain, bulging tympanic membrane). The next step is to discontinue the amoxicillin and start the child on a second-line antibiotic such as amoxicillin–clavulanate (Augmentin) twice a day × 10 days.
- 20. Become knowledgeable about alternative antibiotics for penicillin-allergic patients. If the patient has a gram-positive infection, possible alternatives are macrolides, clindamycin, or quinolones with gram-positive activity such as levofloxacin or moxifloxacin.
- 21. If a patient has an infection that responds well to macrolides, but they think they are “allergic” to erythromycin (symptoms of nausea or gastrointestinal [GI] upset), inform them that they had an adverse reaction, not a true allergic reaction (hives, angioedema).
 - *Example:* Switch the patient from erythromycin to azithromycin (usually a Z-Pack).
- 22. If a patient fails to respond to the initial medication, add another medication (follow the steps of the treatment guideline).
 - *Example:* A patient with chronic obstructive pulmonary disease (COPD) is prescribed ipratropium bromide (Atrovent) for dyspnea. On follow-up, the patient complains that the symptoms are not relieved. The next step would be to prescribe an albuterol inhaler (Ventolin) or a combination inhaler.
- 23. Disease states are usually presented in their full-blown “classic” textbook presentations.
 - *Example:* In a case of acute mononucleosis, the patient will most likely be a teen presenting with the classic triad of sore throat, prolonged fatigue, and enlarged cervical nodes. If the patient is older, but has the same signs and symptoms, it is still mononucleosis (reactivated type).
- 24. Ethnic background may give a clue for some diseases.
 - *Example:* Alpha thalassemia is more common among Southeast Asians (such as Filipinos). Beta thalassemia is more common in Mediterranean people.

25. No asymptomatic or “borderline” cases of disease states are presented in the test.
 - *Example:* In real life, most patients with iron-deficiency anemia are asymptomatic and do not have either pica or spoon-shaped nails. In the exam, they will probably have these clinical findings plus the other findings of anemia.
26. Become familiar with lupus or systemic lupus erythematosus (SLE).
 - *Example:* A malar rash (butterfly rash) is present in most patients with lupus. These patients should be advised to avoid or minimize sunlight exposure (photosensitivity).
27. Become familiar with polymyalgia rheumatica (PMR).
 - *Example:* First-line treatment for PMR includes long-term steroids. Long-term, low-dosed steroids are commonly used to control symptoms (pain, severe stiffness in shoulders and hip girdle). PMR patients are also at higher risk for temporal arteritis.
28. The gold-standard exam for temporal arteritis is a biopsy of the temporal artery. Refer the patient to an ophthalmologist for management.
29. Learn the disorders for which maneuvers are used and what a positive report signifies.
 - *Example:* Finkelstein’s test—positive in De Quervain’s tenosynovitis. Anterior drawer maneuver and Lachman maneuver—positive if anterior cruciate ligament (ACL) of the knee is damaged. The knee may also be unstable. McMurray’s sign—positive in meniscus injuries of the knee.
30. Some conditions need to be evaluated with a radiologic test.
 - *Example:* Damaged joints: Order an x-ray first (but MRI is the gold standard).
31. The abnormal eye findings in diabetes (diabetic retinopathy) and hypertension (hypertensive retinopathy) should be memorized. Learn to distinguish each one.
 - *Example:* Diabetic retinopathy: neovascularization, cotton wool spots, and microaneurysms. Hypertensive retinopathy: arterioventricular (AV) nicking, silver and/or copper wire arterioles.
32. Become knowledgeable about physical exam “normal” and “abnormal” findings.
 - *Example:* When checking deep tendon reflexes (DTRs) in a patient with severe sciatica or diabetic peripheral neuropathy, the ankle jerk reflex (Achilles reflex) may be absent or hypoactive. Scoring: absent (0), hypoactive (1), normal (2), hyperactive (3), and clonus (4).
33. There are only a few questions on benign or physiologic variants.
 - *Example:* A benign S4 heart sound may be auscultated in some elderly patients. A geographic tongue, torus palatinus, and fishtail uvula may be seen during the oral exam in a few patients.
34. Some commonly used drugs have rare (but potentially life-threatening) adverse effects.
 - *Example:* A rare but serious adverse effect of angiotensin-converting enzyme inhibitors (ACEIs) is angioedema. A common side effect of ACEIs is a dry cough (up to 10%).
35. Learn about the preferred and/or first-line drug used to treat some diseases.
 - *Example:* ACEIs or angiotensin receptor blockers (ARBs) are the preferred drugs to treat hypertension in diabetics and patients with renal disease because of their renal-protective properties.
36. When medications are used in the answer options, they will be listed either by name (generic and brand name) or by drug class alone.
 - *Example:* Instead of using the generic/brand name of ipratropium (Atrovent), it may be listed as a drug class (an anticholinergic).
37. Most of the drugs mentioned in the exam are the well-recognized drugs.
 - *Examples:* Additional examples are listed in Chapters 2 and 3.
 - *Penicillin:* Amoxicillin (broad-spectrum penicillin), penicillin VK
 - *Macrolide:* Erythromycin, azithromycin (Z-Pack), or clarithromycin (Biaxin)
 - *Cephalosporins:* First-generation (Keflex), second-generation (Cefaclor, Cefitin, Cefzil), third-generation (Rocephin, Suprax, Omnicef)
 - *Quinolones:* Ciprofloxacin (Cipro), ofloxacin (Floxin)
 - *Quinolones with gram-positive coverage:* Levofloxacin (Levaquin), moxifloxacin (Avelox), gatifloxacin (Tequin)
 - *Sulfa:* Trimethoprim-sulfamethoxazole (Bactrim, Septra), nitrofurantoin (Macrobid)
 - *Tetracyclines:* Tetracycline, doxycycline, minocycline (Minocin)
 - *Nonsteroidal anti-inflammatory drug (NSAID):* Ibuprofen, naproxen (Aleve, Anaprox)
 - *COX-2 inhibitor:* Celecoxib (Celebrex)

38. Category B drugs are allowed for pregnant or lactating women.
 - *Example:* For pain relief, pick acetaminophen (Tylenol) instead of NSAIDs such as ibuprofen (Advil) or naproxen (Aleve, Anaprox). Avoid nitrofurantoin and sulfa drugs during the third trimester (these increase risk of hyperbilirubinemia).
39. The preferred treatment for cutaneous anthrax is ciprofloxacin 500 mg orally twice a day for 60 days or 8 weeks. If the patient is allergic to ciprofloxacin, use doxycycline 100 mg twice a day. Cutaneous anthrax is not contagious; it comes from touching fur or animal skins that are contaminated with anthrax spores.
40. Approximately 15% of the questions will be about infants and children.
 - *Example:* The American Academy of Pediatrics (AAP) recommends that breastfed infants should be started on vitamin D during the first few days of life; then at age 4 months, iron-supplementation is recommended (formula contains vitamins/minerals so no need to supplement).
41. A common question design regarding pediatrics is to query about growth and development norms (or abnormal).
 - *Example:* At the age of 6 months, infants can sit without support; roll over front to back, then from back to front; transfer objects from one hand to the other; use a raking grasp; and babble.
42. Physical exam findings in children that are abnormal with serious consequences might be included.
 - *Example:* Leukocoria (white color) is noted on one eye while checking for the red reflex. Rule out retinoblastoma of the eye, which is a malignant tumor of the retina.
43. There are some questions on theories and conceptual models.
 - *Example:* Stages of change or “decision” theory (Prochaska) include concepts such as precontemplation, contemplation, preparation, action, and maintenance (see Chapter 26).
44. Other health theorists who have been included on the exams in the past are (not inclusive) Alfred Bandura (self-efficacy), Erik Erikson, Sigmund Freud, Elisabeth Kübler-Ross (grieving), and others.
 - *Example:* If a small child expresses a desire to marry a parent of the opposite sex, the child is in the oedipal stage (Freud). The child’s age is about 5 to 6 years (preschool to kindergarten).
45. Starting at the age of about 11 years, most children can understand abstract concepts (early abstract thinking) and are better at logical thinking.
 - *Example:* When performing the Mini-Mental State Exam, when the NP is asking about “proverbs,” the nurse is assessing the patient’s ability to understand abstract concepts.
46. *Keep these good communication rules in mind:* Ask open-ended questions; do not reassure patients, avoid angering them, and respect their culture. There may be two or three questions relating to “abuse” (e.g., child abuse, domestic abuse, elderly abuse).
47. Follow national treatment guidelines for certain disorders. Following is a list of treatment guidelines used as references by the ANCC and the AANPCB.*
 - **Asthma:** Global Initiative for Asthma. (2020). 2020 GINA Report, Global Strategy for Asthma Management and Prevention. <https://ginasthma.org/gina-reports/>.
 - **COPD:** Global Initiative for Chronic Obstructive Lung Disease. (2020). *Global strategy for the diagnosis, management, and prevention of chronic obstructive pulmonary disease, 2020*.
 - **Diabetes:** American Diabetes Association Standards of medical care in diabetes. (2020). *Diabetes Care*, 43(Suppl. 1), S14–S31.
 - **Ethics:** Fowler, M. (2015). *Guide to the code of ethics for nurses with interpretive statements: Development, interpretation, and application* (2nd ed.). Nursesbooks.org.
 - **Healthy People:** Office of Disease Promotion and Health Prevention. (2010). *Healthy People 2030*. <https://health.gov/healthypeople/objectives-and-data/browse-objectives>.
 - **Health Promotion:** U.S. Preventive Services Task Force. www.uspreventiveservicestaskforce.org/uspstf/.
 - **Hyperlipidemia:** 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: A Report of the ACC/AHA Task Force on Practice Guidelines. *Circulation*, 139(25).
 - **Hypertension:** 2017 Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. (2018). *Journal of the American College of Cardiology*, 71, e127–e248.

* Not an all-inclusive list.

- **Mental Health:** American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). American Psychiatric Publishing.
 - **Pediatrics:** American Academy of Pediatrics. (2017). *Bright futures: Guidelines for health supervision of infants, children, and adolescents* (E-book; 4th ed.).
 - **STDs (or Sexually Transmitted Infections):** Centers for Disease Control and Prevention. (2015). *Sexually transmitted diseases treatment guidelines*. U.S. Department of Health and Human Services.
48. The ANCC exam has several questions about evidence-based medicine that are designed in the drag-and-drop format. For important facts, review Chapter 2 to become familiar with this format, as well as Chapter 30 for important facts.
 49. If the question is asking for the initial or screening lab test, it will probably be a “cheap” and readily available test such as the complete blood count (CBC) to screen for anemia.

MAXIMIZING YOUR SCORE

1. The first answer that “pops” in your head is usually the correct answer.
2. If you are guessing, pick an answer at random. Try not to change it unless you are sure.
3. Use the “flag” command so that you can return to the question later when you complete the exam.
4. Avoid changing too many answers. If you are not sure, then leave the answer alone. I advise my students not to change more than three answers.
5. Avoid choosing exotic diseases as answers if you are guessing. Remember, these are tests for primary care conditions.
6. One method of guessing is to look for a pattern. Pick the answer that does not fit the pattern. Another is to pick the answer that you are most “attracted” to. Go with your gut feeling, and do not change the answer unless you are very sure.
7. Remind yourself to read slowly and carefully throughout the test. Avoid reading questions too rapidly. Make sure that you understand the stem of the question (usually the last sentence of a question).
8. If you are having problems choosing or understanding the answer options, try to read them from the bottom up (from option D to A, or from 4 to 1).
9. Eliminate the wrong answers after you have read all the answer options. If an answer option contains all-inclusive words (*all, none, every, never, none*), then it is probably wrong.
10. Be careful with certain words such as *always, exactly, often, sometimes, and mostly*.
11. Assume that each question has enough information to answer it correctly. Questions and answers are carefully designed. Take the facts at face value.
12. The first few questions are usually harder to solve. This is a common test design. Do not let it shake your confidence. (Guess the answer and “mark” it so that you can return to it later.)
13. Save yourself time (and mental strain) by reading the last sentence (or stem) of long questions and case scenarios *first*. Then read the question again from the beginning. The advantage of this “backward reading” technique is that you know ahead of time what the question is asking for. When you read it again “normally,” it becomes easier to recognize important clues that will help you answer the question.
14. Just because a statement or an answer is true does not mean that it is the correct answer. If it does not answer the stem of the question, then it is the wrong answer.
15. Design and memorize your “scratch paper” a few weeks before you take the exam. Choose what you want on it wisely. Remember to keep it brief.
16. Use the time left from the “free” computer tutorial time to write down the facts that you memorized for your scratch paper. If you run out of time, skip this step.
17. Some suggestions of facts to write down on the scratch paper are lab results (e.g., hemoglobin, hematocrit, MCV, platelets, WBC count, neutrophil percentage, potassium, urinalysis). Other popular choices are the murmurs, mnemonics, and cranial nerves.
18. Do not leave any of the questions unanswered, because there is no penalty for guessing. Questions that are left blank are marked as errors (0 points). If you have only 30 minutes left and you are not done, quickly answer the remaining questions at random.
19. If you spend more than 60 seconds on a question, you are wasting time. Answer it at random and then “mark” it so that you can return to it later (after you finish the entire test).
20. Consider a quick break (if you have enough time) if you get too mentally fatigued. Solving 150 to 175 questions is pretty intense. If you feel “fuzzy” or tired, go to the bathroom and get a drink of

water, and splash cold water on your face. This can take <5 minutes. You can bring bottled water, but you have to leave it outside the testing area.

21. The countdown clock in the computer does not stop for breaks. Do not use more than 5 minutes for your quick bathroom break.
22. If you have failed the test before, try not to memorize what you did on the previous exam you took. The answers you remember may be wrong. Pretend that you have never seen the test before so that you can start out fresh mentally.
23. Do not panic or let your anxiety take over. Learn to use a calming technique, such as deep breathing, to calm yourself quickly before you take the exam.
24. Consider using a test-anxiety hypnosis CD and use it every night for at least 2 to 3 weeks for maximum effect.
25. One of the most important pieces of advice that I can give you is to make sure that you get enough sleep the night before the exam. Aim for at least 7 to 8 hours of sleep. Better yet, make sure that you get enough sleep a few nights before, not just the night before.
26. Before the test, practice answering questions and check the rationales afterward. The ANCC has free questions per specialty that you can take online and score. Buy practice questions from the ANCC or the AANPCB (depending on which test you are taking).
27. Check out the website allnurses.com and search for examination tips from others who have taken/failed the FNP certification exam. You can also do this directly by using a search engine such as Google.

PREPARING FOR THE EXAM

Review Timeline

1. Start seriously reviewing for the exam at least 3 months in advance. Study time can range from 2 to 3 hours per session, or you can break it up into 1-hour segments. Be consistent. The best time to study is the time of day you are most alert.
2. Prepare a study schedule by organ system. Photocopy the table of contents of your primary care textbook. Place check marks next to the diseases that you want to concentrate on and then schedule the date and time period for each organ system.
3. Concentrate your studies on your weaker areas. For example, if orthopedics is one of your weak areas, devote more time to it. Spend 3 days on orthopedics versus 1 day for the “easy” organ systems (since you understand them better).
4. I highly recommend that you attend at least one quality review course and buy CDs of another review course. I teach review courses live and by webinar. You can view my class schedule at my website at www.npreview.com.
5. Buy a new notebook. If you find that you are having a problem understanding a concept, write it down in your notebook, then research it and find the answer to your question.
6. Meet with someone in your local area or by phone who is also taking the exam. Practice quizzing each other.
7. Note the disease and topics that I have highlighted in this review book. Some organ systems have more “weight” on the exam than others. Become more familiar with these areas.
8. If you are taking the ANCC exam, I highly recommend that you devote at least 30% of your study time learning about the nonclinical topics.
9. If you learn better in a group, organize one. Decide ahead of time what organ systems or diseases to cover together so that you do not waste time.

Testing Center Details

After closing briefly due to COVID-19, testing has resumed in many U.S. and Canada test centers, either at full or limited occupancy (based on local and government mandates). While in the testing center, you must wear a mask at all times. Your temperature will be checked. If you are febrile or have COVID-like symptoms, call the testing center for advice on how to reschedule your exam.

1. Call the testing center or confirm your appointment online 4 weeks before to verify your appointment date and time.
2. Locate and drive to the testing center 1 to 2 weeks before taking the exam. Save the address of the testing center on your GPS or map app.

3. Arrive at least 45 to 60 minutes before your scheduled time so that you have enough time to park your car, locate the testing center, and check in for your appointment. Do not forget to bring your mask to the testing center.
4. Carry acceptable forms of unexpired primary ID (photo with signature). If you do not have the proper forms of identification, you will not be allowed to take the exam.
 - Driver's license issued by the Department of Motor Vehicles (DMV)
 - State ID card issued by the DMV
 - Passport
 - U.S. military ID
 - *Your secondary ID (nonphoto)*: Card with your name and signature (e.g., credit card, ATM card, Social Security card, voter registration card, employee ID)
 - Expired IDs are not acceptable.
 - The name that is printed on your primary ID must match the name that you used when you applied for the exam.
 - Check that both your primary and secondary IDs and the letter with your Prometric identification number match; make sure that all your IDs are inside your wallet/handbag.
5. Biometrics are used for enhanced security. The test-taking room is also monitored closely by videotape and microphones. Your glasses will be checked visually, and you will be asked to show your arms and ankles, as well as empty your pockets from an agreed-upon safe distance.
6. A whiteboard (8.5 × 11 inches) with a marker, or scratch paper and a pencil, is given to you by the testing center staff (and collected after you are done). If you are given paper and tend to write a lot, ask for extra paper.
7. You can request noise-reducing headsets or earplugs; consider this option if you are sensitive to noise.
8. Cell phones, watches, cameras, pagers, jewelry (except for engagement and/or wedding ring), and food or drink are not allowed inside the testing area.
9. Testing computers are predetermined. Each test-taker is assigned one small cubicle with one computer. If you are having problems seeing the computer screen, bring it to the proctor's attention as soon as possible.
10. Verify that you are given the correct examination as soon as you sit down. Check that the title and the examination code on the screen match the information sent by the testing agency for your NP specialty.
11. No food or drink is allowed inside the testing room. But you can bring in a drink and food (e.g., snack bar) and place it outside the locker (on top of the locker) so that you do not have to open your locker. You are not permitted to unlock your locker during the testing period.
12. If you need to go to the restroom, you must first sign out with the proctor/testing staff. Remember, the time clock will continue to count down the time. It does not stop for breaks.
13. Do not forget your glasses if you need them to read text on the computer.

The Night Before the Exam

1. Avoid eating a heavy meal or consuming alcoholic drinks the night before the exam. Avoid eating 3 to 4 hours before bedtime (prevents heartburn).
2. Get enough sleep. Aim for at least 7 to 8 hours. Getting adequate sleep is probably one of the most important things you can do to help you pass the exam.
3. If you are scheduled to take your exam in the morning, set two alarms to wake you up on time. Give yourself extra time if it is a weekday (traffic congestion).

The Day of the Exam

1. Avoid eating a heavy breakfast or eating only simple carbs. The best meals are a combination of a protein with a complex carbohydrate (e.g., eggs, whole-wheat bread, nuts, cheese). If you eat only simple carbs, you can become hypoglycemic within 1 hour.
2. If you get drowsy and "fuzzy" during the exam, there are several ways to "wake up" rapidly. Excuse yourself and go the bathroom to splash cold water on your face. You can perform 10 to 20 "jumping jacks." Drinking cold water, a caffeinated beverage, or coffee can be very helpful. (However, don't drink too much.)
3. Avoid drinking too much fluid and do not forget to empty your bladder before the exam.

4. Wear comfortable clothing and dress in layers in case of temperature changes.
5. Jewelry is not allowed inside the testing area, except for wedding and/or engagement rings.

OTHER TEST-TAKING ISSUES

Test Anxiety

It is normal to feel anxious before taking an exam. A little anxiety helps us to become alert and vigilant, but too much anxiety can wear you down both emotionally and physically.

Your internal perception about how well you will do on the exam is very important. If you tend to become very anxious (or you are now more anxious because you previously failed the exam), there are calming methods that may prove helpful in controlling your anxiety. Following are some suggestions to help reduce test-taking anxiety a few weeks before the exam.

1. Make sure that you have devoted enough time for your review studies. If you feel in your gut that you have not studied “enough” or that you are not ready, it will worsen anxiety.
2. Consider signing up for both the ANCC and AANPCB exams. It can help to reduce your anxiety level because you feel like you have a “backup” exam.
3. If you have a firm job offer, I recommend that you sign up for both the exams to decrease test-taking anxiety.
4. Consider taking one or two review courses. My review courses and webinars are listed at my website at www.npreview.com.
5. Avoid negative “self-talk.” When you catch yourself doing it, silently tell yourself to “Stop it! Right now!”
6. Improve the nutritional content of your diet, especially about 4 weeks before the exam. Vitamins and supplements that may be helpful to the brain include omega-3 capsules (fish oil); B-complex vitamins; vitamins C, D, and E; and choline.
7. Hypnotherapy audio files are sold online for use in portable-player devices. For maximum effectiveness, it is best to listen to them daily for 2 to 4 weeks.
8. “Tapping” is another method used to counteract test anxiety (or increase self-confidence). The technique is demonstrated in videos posted on YouTube.

Your Panic Button

If you find yourself feeling panicky during the exam, try the following calming technique. Practice this exercise at home until you feel comfortable doing it.

Breathing Exercise

1. Close your eyes. During inhalation, tell yourself gently, “I am breathing in calmness.” Then hold your breath and count from 1 to 3.
2. During exhalation, tell yourself gently, “I am breathing out fear,” while counting from 1 to 5.
3. Inhale deeply through your nose and exhale slowly through your mouth.
4. Complete one cycle of three inhalations and three exhalations. Repeat as needed.

Question Dissection and Analysis

2

EVIDENCE-BASED MEDICINE

I. Discussion

Questions about evidence-based medicine (EBM) are typically designed in “drag-and-drop” format on the American Nurses Credentialing Center (ANCC) exam. They are presented as two boxes with three sections each. The example that follows illustrates this format. On the left side, three types of article summaries (or research studies) are marked A, B, and C (in the exam, they will appear as blue boxes). On the right side, you will see the rankings from 1 to 3 (in the exam, they will be yellow boxes). You “drag” one of the articles from the left-hand box into the correct ranking or hierarchy in the right-hand box. Your job is to rank the three research article summaries as 1, 2, or 3 (best evidence, moderate evidence, weakest evidence).

II. Example

Article Summaries	Strength of Evidence (strongest to weakest)
A. A meta-analysis on MEDLINE and Cochrane databases that found 53 RCTs about ginkgo biloba use in patients with early dementia	1.
B. A specialty society opinion paper regarding the effectiveness of ginkgo biloba supplementation in dementia	2.
C. An experimental study on 500 patients with early dementia who were given ginkgo biloba daily for 6 months versus the control group, which was given placebo pills	3.

RCTs, randomized controlled trials.

III. Correct Answer

Article A is ranked 1, Article C is ranked 2, and Article B is ranked 3.

IV. Question Dissection

Best Clues

I teach a three-step system to students in my review courses:

- First, identify the research study that has the highest/best level of evidence (#1 ranking). An easy way to do this is by searching for keywords in the question such as *meta-analysis*, *systematic review*, or *randomized controlled trials* (RCTs) combined with *Cochrane Database of Systematic Reviews* (CDSR), MEDLINE® database, and/or Cumulative Index to Nursing and Allied Health Literature (CINAHL®).
- Next, look for the research study that has the weakest evidence (#3 ranking). It usually contains keywords such as *expert opinion* or *specialty society*.
- Therefore, view the study that is leftover as having the middle ranking (#2 ranking). In addition, keep in mind that if the choices do not include a meta-analysis or systemic review, then the study with the highest ranking would be an RCT or an experimental study. The ANCC exams usually have several of this type of question. Further discussion about EBM appears in Chapter 30.
- The levels of evidence rankings are:
 - Meta-analysis and/or systematic reviews (Cochrane/MEDLINE/CINAHL/PubMed)

- RCTs (used for testing medical treatment effectiveness, subjects assigned at random to either a control or treatment group)
- Experimental studies (control group, intervention group, randomization)
- Cohort/case control studies
- Retrospective chart reviews
- Expert/specialty society opinions

PHOTOGRAPHS

I. Discussion

On the ANCC exam, expect to see several color photographs of skin conditions and a few on the eyes/fundi. In the future, there may be pictures from other organ systems. The stem of the question usually will ask for the possible diagnosis, differential diagnosis, or type of treatment. If you plan to take the ANCC exam, you need to memorize how a skin condition or eye finding appears in a color photo. It is a good idea to use a search engine (e.g., Google) to look for the images. For example, you want to become familiar with skin cancers, such as basal cell carcinoma and melanoma, and fundoscopic findings in diabetes and hypertension.

II. Example

The nurse practitioner (NP) is performing a routine physical exam on a 54-year-old White male farmer who is an immigrant from Australia. The NP notices a round skin lesion on the patient's head. It has a firm texture with indurated edges and telangiectasia (Figure 2.1). The patient reports that the lesion does not itch, but it has slowly enlarged over the past few years. Which of the following conditions is most likely in this patient?

- A. Nodular melanoma
- B. Squamous cell carcinoma
- C. Basal cell carcinoma
- D. Actinic keratosis



Figure 2.1 Sample exam photograph.

Source: National Cancer Institute

III. Correct Answer: Option C

C. Basal cell carcinoma

IV. Question Dissection

Best Clues

- Notice that the skin lesion has a pearly or waxlike (shiny) appearance with telangiectasia, which is “classic” for basal cell carcinoma; some lesions may show central ulceration.
- Patient has risk factors for skin cancer, such as light-colored skin, and he is from Australia, which has high rates of skin cancer.

- The skin lesion is located on a sun-exposed area.
- It is probably not nodular melanoma, which usually has pigment such as brown or black color with irregular borders.
- Actinic keratosis is a precancer of squamous cell carcinoma and is usually located on the scalp (males), face, and back of the hands (dorsum); they appear as a crusty/scaly growth that slowly enlarges over time.
- The gold-standard test for skin cancer is the skin biopsy.

MULTIPLE-CHOICE QUESTIONS WITH MORE THAN ONE CORRECT RESPONSE

I. Discussion

Expect to see some multiple-choice questions with five or six answer options in both the ANCC and American Academy of Nurse Practitioners Certification Board (AANPCB) exams. The question will ask for two or three correct answers. For example, you may be asked to pick two or three differential diagnoses for a case of skin rash. The clues are given in the presentation of the signs and symptoms.

II. Example

A 75-year-old woman with mild dementia, hyperlipidemia, and emphysema is brought in by her middle-aged daughter as a walk-in patient in a community clinic with a complaint of a sudden onset of red rashes on her left lower arm and hand. During the skin exam, the NP notes that there are a few blisters. When the NP touches one of the blisters, it ruptures and drains clear serous fluid. Which of the following three conditions should the NP consider in the differential diagnosis?

- A. Contact dermatitis
- B. Erysipelas
- C. Psoriasis
- D. Impetigo
- E. Thermal burn

III. Correct Answer: Options A, D, and E

- A. Contact dermatitis
- D. Impetigo
- E. Thermal burn

IV. Question Dissection

Best Clues

- Easily ruptured blisters (fragile) is a classic finding for bullous impetigo, an acute bacterial skin infection caused by *Staphylococcus* or *Streptococcus*.
- Contact dermatitis can present with just red skin or red skin with blisters. The rash can be located anywhere on the body, and it may have a pattern (like a belt) or no pattern.
- The timing of the rash is very important. Is it acute or chronic? Rule out option C (psoriasis), which is a chronic skin disease.
- Erysipelas is a type of cellulitis caused by strep. It resembles a bright-red, warm, raised rash (plaque-like) with discrete borders usually located on the face or the shins. Blistering is not present.
- A thermal burn is a burn caused by heat (e.g., fire, heat). Consider a second-degree burn in the differential diagnosis because of its acute onset. Also, the patient has mild dementia, which puts her at a higher risk for accidents.

DIAGNOSTIC IMAGING

I. Discussion

Questions about diagnostic imaging tests may appear on the exam. You may get a multiple-choice question alone or a question that is accompanied by a chest x-ray film. I recommend that you use a search engine (e.g., Google) to search for images of chest films with lobar consolidation due to community-acquired pneumonia (CAP), right middle lobe pneumonia, pulmonary tuberculosis (TB) infection, and emphysema/chronic obstructive pulmonary disease (COPD).

II. Example

A 34-year-old male smoker presents in an urgent care clinic complaining of a productive cough, chest congestion, fever, chills, and poor appetite for 1 week. Cough is productive of greenish sputum, which is sometimes tinged with a small amount of blood. Vital signs are temperature of 101.2°F, pulse of 100 beats/min, respirations of 24 breaths/min, and blood pressure (BP) of 122/88 mmHg. A radiograph of the chest is obtained (Figure 2.2). What is the most likely diagnosis in this patient?

- A. Acute bronchitis
- B. Right middle lobe pneumonia
- C. Right lower lobe pneumonia
- D. Legionella pneumonia

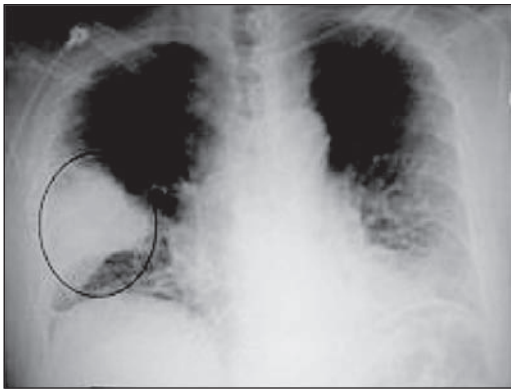


Figure 2.2 Sample exam photo.

III. Correct Answer: Option B

- B. Right middle lobe pneumonia

IV. Question Dissection

Best Clues

- The signs and symptoms of CAP include fever and cough productive of green sputum with a small amount of blood (or rust-colored sputum). Rust-colored sputum is associated with *Streptococcus pneumoniae* infection.
- Most of the middle lobe of the right lung is in the anterior chest by the right nipple area. Notice that lobar consolidation is located in the same area.
- Patients with acute bronchitis may have chest congestion but not fever, chills, or productive cough with purulent sputum.
- Legionella pneumonia (Legionnaire's disease) is uncommon in primary care. Look for a history of exposure to "nebulized" water sources (e.g., air conditioners, fountains). Presents with pneumonia signs/symptoms that are accompanied by gastrointestinal (GI) symptoms (diarrhea, nausea/vomiting).

CULTURE

I. Discussion

There will be several questions on the ANCC exam that address culture. The questions will address knowledge of cultural practices that influence health-seeking behavior. Some of the cultures that may be included in the ANCC exam are Hispanic/Latino, Muslim, Chinese, Native American (Navajo), and Southeast Asian (Vietnamese, Hmong, Filipinos, and others).

II. Example

An elderly woman immigrant from Vietnam who recently has been diagnosed with hypertension is returning for a 4-week follow-up visit. The patient is on a prescription of hydrochlorothiazide 12.5 mg daily. Her BP during the visit is 150/94 mmHg. The NP queries the patient about whether she is taking her medication. The patient looks down at the floor and does not directly answer the question. Which of the following statements regarding the health behaviors of Southeast Asians is incorrect?

- A. The patient may have difficulty verbalizing questions about their treatment.
- B. The patient may ask to consult with an older family member about major health decisions.
- C. If the patient is not compliant with taking medications, they will not directly communicate it to the health provider.
- D. The patient will directly verbalize their disagreement in a loud voice.

III. Correct Answer: Option D

D. The patient will directly verbalize their disagreement in a loud voice.

IV. Question Dissection

Best Clues

- The patient's behavior (aggressive, loud) is considered rude in most Asian cultures.

NOTES

1. The Hmong ethnic group that immigrated to the United States came from several countries (e.g., Laos, Vietnam, Thailand). A traditional household has a male (i.e., father) who is the head of the household. The family identifies with a clan group, which is headed by an older male.
2. For major health decisions, the head of the family is always involved. If the father is dead, then another older male relative (e.g., uncle) may be consulted.
3. Most Asians have high regard for physicians (and college education). Because of this cultural value, they may not directly disagree or question the healthcare provider.
4. Asian Americans and Pacific Islanders are 8 to 13 times more likely to develop liver cancer than other groups because of higher rates of chronic hepatitis B infection.

LAB RESULTS AND DIAGNOSTIC TESTS

I. Discussion

Laboratory tests, such as hemoglobin and hematocrit, mean corpuscular volume (MCV), total white blood cell (WBC) count, percentage of neutrophils in the WBC differential, thyroid-stimulating hormone (TSH), prostate-specific antigen (PSA), and urinalysis (UA), are commonly encountered in the exams. Learn the significance of the abnormal results and the follow-up tests that are needed to evaluate them further.

The AANPCB exam does list the norms for some of the common laboratory tests. They will appear only when needed to answer a question (such as an anemia question).

In contrast, the ANCC does not list any of the normal results in its certification exams. Therefore, if you plan to take the ANCC exam, it is important that you memorize the normal results of these laboratory tests.

Be warned that lab results are also used as distractors; the labs listed may not be necessary to solve the exam question correctly. The normal results for these labs are also included in the pertinent review chapters of this book (e.g., TSH will be found in Chapter 9).

II. Example

An elderly man of Mediterranean descent has a routine complete blood count (CBC) done for an annual physical. The following are his lab test results: hemoglobin is 12.0 g/dL, hematocrit is 39%, and MCV is 72 fL. His PSA result is 3.2 ng/mL. UA shows no leukocytes and few epithelial cells. Which of the following laboratory tests are indicated for this patient?

- A. Serum iron, serum ferritin, total iron-binding capacity (TIBC), and the red cell distribution width (RDW)
- B. Serum vitamin B12 and folate level with a peripheral smear
- C. CBC with white cell differential and UA
- D. Urine culture and sensitivity with microscopic exam of the urine (Tables 2.1 and 2.2)

III. Correct Answer: Option A

- A. Serum iron, serum ferritin, total iron-binding capacity (TIBC), and the red cell distribution width (RDW)

Table 2.1 List of Laboratory Norms (Adults)

CBC	Reference Ranges
Hemoglobin	
Males	13.0–17.5 g/dL
Females	12.0–16 g/dL
Hematocrit	
Males	40%–50%
Females	36%–45%
MCV	80–100 fL
RDW	>14.5%
Platelet count	<150,000/mm ³ (increased risk of bleeding, disseminated intravascular coagulation)
Reticulocytes	0.5%–1.5% of red cells (↑ acute bleeding), starting treatment for vitamin deficiencies (iron, B12, folate), acute hemolytic episodes
Total WBC count	4,500–11,000/mm ³ (↑ bacterial infections)
Neutrophils (or segs)	55%–70% (↑ bacterial infections)
Band forms (immature WBCs)	>5% (↑ severe bacterial infections) Also called "shift to the left"
Eosinophils	>3% (↑ allergies, parasitic diseases, cancer)

CBC, complete blood count; MCV, mean corpuscular volume; RDW, red cell distribution width; WBC, white blood cell.

Table 2.2 List of Blood Chemistries

Laboratory Test	Reference Ranges
TSH	>5.0 mU/L (hypothyroidism) <0.4 mU/L (hyperthyroidism)
PSA	<4.0 ng/mL (benign prostatic hyperplasia [BPH], prostate cancer)
Ferritin	<15 mcg/L (iron-deficiency anemia)
ESR; sed rate	Men 0–22 mm/hr Women 0–29 mm/hr Elevated (giant cell arteritis, rheumatoid arthritis [RA], lupus, inflammation)
CRP	Elevated (inflammation, autoimmune diseases, a risk factor for heart disease)
cTnT	Elevated in myocardial infarction, heart damage, heart failure Sensitive test for myocardial cell damage (myocardial infarction, unstable angina)
BNP	Elevated (elevated in heart failure)
Potassium	<3.5 or >5.5 mEq/L (critical values <2.5 or >8 mEq/L)

BNP, B-type natriuretic peptide; CRP, c-reactive protein; cTnT, cardiac troponins; ESR, erythrocyte sedimentation rate; PSA, prostate-specific antigen; TSH, thyroid-stimulating hormone.

IV. Question Dissection

Best Clues

- Low hemoglobin and hematocrit for gender (male) and age (abnormal CBC result)
- An MCV of 72 fL, which is indicative of microcytic anemia (assessment)
- The ethnic background of the patient (demographics)
- Ignore the UA and PSA tests because they are not necessary to solve the problem

NOTES

1. You must go through three steps to answer this question correctly:
 - *First step:* A hemoglobin of <13.5 g/dL in males (but not in females) is indicative of anemia. An MCV of 72 fL is indicative of microcytic anemia (norm 80–100 fL).
 - *Second step:* The MCV will direct you in the differential diagnosis (microcytic, normocytic, or macrocytic).
 - *Third step:* The differential diagnosis for microcytic anemia is iron deficiency and alpha or beta thalassemia trait or minor for the exams.
2. In iron-deficiency anemia, the following results are found:
 - *Serum ferritin and serum iron levels:* Decreased
 - *TIBC and RDW:* Elevated
3. In alpha or beta thalassemia trait or minor, the following results are found:
 - *Serum ferritin and serum iron levels:* Normal
 - *TIBC:* Normal
4. The gold-standard test to diagnose any anemia involving abnormal hemoglobin (e.g., thalassemia, sickle cell) is the hemoglobin electrophoresis.
5. The RDW is a measure of the variability in size of red blood cells (RBCs; or anisocytosis). An elevated RDW is one of the earliest indicators of iron-deficiency anemia.
6. In clinical practice, rule out iron-deficiency anemia first (most common anemia in the world for all ages/races/gender) before ordering a hemoglobin electrophoresis.

PHARMACOLOGIC CONCEPTS

I. Discussion

The current version of the exams includes more topics about pharmacology. To solve a question, you need to understand key terms, their definitions, and how they are applied. There are no math problems where you must figure out the correct dose of a drug. See Chapter 3 for further discussion.

II. Examples

- *Deprescribing:* The process of adjusting the dose of a medication down to the minimum effective dosage or stopping it when it is no longer needed (or if the patient's health status has changed). Deprescribing helps to decrease polypharmacy, especially in the geriatric population. For example, there are evidence-based deprescribing guidelines for proton-pump inhibitors and benzodiazepines, which can be found online.
- *Pharmacogenomics/pharmacogenetics:* Your genes affect how you respond (or do not respond) to a drug. It is now known that certain genes affect how the body metabolizes certain drugs. It allows us to truly "individualize" medicine.
- *Pharmacokinetics:* The process by which a medication is absorbed, distributed, metabolized, and eliminated by the body.
- *Half-life:* Amount of time it takes for a drug concentration to reduce to half of its original dose.
- *First-pass metabolism (first-pass effect):* When an oral drug reaches the stomach, it goes to the small intestines, then to the portal circulation where it is metabolized by the liver. After the liver has metabolized the drug, it is released to the systemic circulation where it can reach the target organ(s).

WOMEN'S HEALTH

I. Discussion

Expect to see several questions addressing gynecological conditions. The topics that are covered include recognizing and treating sexually transmitted infections or diseases (STIs or STDs), oral contraceptive issues, abnormal Pap smears, menopausal conditions, threatened abortion or early pregnancy, and many more. For example, there will be questions about vaginal disorders, such as bacterial vaginosis (BV), candida vaginitis, trichomoniasis, and atrophic vaginitis. In pregnant patients, use only medications that are FDA Category A or B or those that have no adverse effect on pregnancy and lactation.

Be aware that a clinical finding can be described in detail instead of using its common name. For example, the term *clue cell* is not used in the question that follows. Instead, it is described in detail ("mature squamous epithelial cells with numerous bacteria on the cell surface and borders").

II. Example

An 18-year-old female student presents in the college health clinic complaining of a strong odor in her vagina. She reports that she had an abortion about 3 weeks ago and recently completed her prescription of antibiotics. The NP performs a vaginal speculum exam and notes a large amount of gray to off-white discharge coating the patient's vaginal walls. It has a milk-like consistency. During microscopy, the slide reveals mature squamous epithelial cells with numerous bacteria on the cell surface and borders. The vaginal pH is 6.0. Which of the following conditions is most likely?

- A. Trichomoniasis
- B. BV
- C. Candida vulvovaginitis
- D. Hormonal changes

III. Correct Answer: Option B

- B. BV

IV. Question Dissection

Best Clues

- The vaginal pH is alkaline (pH of 6.0).
- Rule out *Candida* because it is classified as a yeast organism (not a bacteria).
- Rule out *Trichomonas* because it is a protozoan or unicellular flagellated organism.
- The odor and discharge are not due to hormonal changes in an 18-year-old.

NOTES

1. BV has an alkaline pH (vagina normally has an acidic pH of 3.5–4.5). BV is the only vaginal condition with an alkaline pH for the exam.
2. BV is not considered an STD (it is caused by an imbalance of vaginal bacteria). The sex partner does not need to be treated. It is a vaginosis (not a vaginitis).
3. BV does not cause inflammation (the vulvovagina will not be red or irritated). The microscopy slide will have very few WBCs and a large number of clue cells.
4. The vaginal discharge in *Candida* infection is a white color with a thick and curdlike consistency. It frequently causes redness and itching in the vulvovagina because of inflammation.
5. The microscopy in candidiasis will show a large number of WBCs, pseudohyphae, and spores ("spaghetti and meatballs").
6. *Candida* yeast is a normal flora of the GI tract and in some women's vaginas.
7. *Trichomonas* infection (or trichomoniasis) vaginal discharge is copious, bubbly, and green in color. It causes a lot of inflammation, resulting in itching and redness of the vulvovagina. It is considered to be an STI. The sex partner also needs treatment.
8. PCR testing is now available for trichomonas and vaginal candida infections. For trichomonas, a urine specimen (males or females) is used. For vaginal candida, a vaginal swab is used.

MEDICATIONS

I. Discussion

When studying pharmacology for the exam, it is generally not important to memorize the specific drug doses. What is more important is to study a drug's "safety" issues, such as contraindications, major drug/food interactions, and well-known side effects.

You will need to be familiar with the drug's indications and the duration of treatment. Become familiar with a "first-line drug" and the alternative drug (if applicable). For example, the first-line (or preferred) drug for treating "strep" throat is still penicillin V PO $\times$ 10 days. If the patient has a penicillin allergy, macrolides and clindamycin can be used instead.

The majority of the medicines seen on the exam are the well-known drugs that have been in use for a few years to many decades (e.g., doxycycline, penicillin, amoxicillin). Memorize drug class and some representative drugs from that class. For example, in the quinolone drug class are drugs such as ofloxacin (Floxin), moxifloxacin (Avelox), and levofloxacin (Levaquin). A drug's brand name may not be used on the exam; instead, only the drug's generic name or drug class may be used. In addition, you need to become familiar with some of the U.S. Food and Drug Administration (FDA) category X drugs (see Chapter 3).

II. Examples

Example A

Using the drug class as the answer option:

A previously healthy 30-year-old complains of an acute onset of fever and chills accompanied by a productive cough with purulent sputum and a loss of appetite. The patient denies receiving an antibiotic in the previous 3 months. The NP diagnoses CAP. The Infectious Diseases Society of America (IDSA) and the American Thoracic Society (ATS) treatment guidelines recommend which of the following as the preferred first-line treatment for this patient?

- A. Macrolides
- B. Antitussives
- C. Cephalosporins
- D. Fluoroquinolones with gram-positive bacteria activity

Example B

Which of the following antibiotics is the preferred treatment for healthy adults diagnosed with uncomplicated CAP?

- A. Azithromycin (Zithromax Z-Pak 250 mg) 500 mg on day 1, then 250 mg daily for 4 days
- B. Dextromethorphan with guaifenesin (Robitussin DM) 1 to 2 teaspoons PO QID as needed
- C. Cephalexin (Keflex) 500 mg PO QID $\times$ 10 days
- D. Levofloxacin (Levaquin) 500 mg PO daily $\times$ 7 days

Example C

The following is an example of a question about a common side effect:

Which of the following are possible side effects that may be seen in a patient who is being treated with hydrochlorothiazide for hypertension?

- A. Dry cough and angioedema
- B. Swollen ankles and headache
- C. Hyperuricemia and hyperglycemia
- D. Fatigue and depression

III. Correct Answers

Example A: Option A

- A. Macrolides

Example B: Option A

A. Azithromycin (Zithromax Z-Pak 250 mg) 500 mg on day 1, then 250 mg daily for 4 days

Example C: Option C

C. Hyperuricemia and hyperglycemia

IV. Question Dissection**Best Clues**

- Lack of comorbidity (“healthy adults”) is an important clue in Examples A and B.
- Your knowledge of the latest IDSA and the ATS treatment guidelines helps to correctly answer Examples A and B (covered in Chapter 8).
- For Example C, you must memorize the adverse side effects of thiazide diuretics.

NOTES

1. According to the IDSA and the ATS treatment guidelines, outpatient treatment of CAP in healthy patients (no comorbidities) involves the macrolides (azithromycin, clarithromycin, or erythromycin).
2. Example A is caused by angiotensin-converting enzyme (ACE) inhibitors. Look for a sudden or new onset of a dry cough in a patient with hypertension (without signs of the common cold). Angioedema is a rare adverse effect and can be life-threatening.
3. Example B is caused by calcium channel blockers (CCBs). Look for a hypertensive patient with swollen ankles (not associated with heart failure) and headache.
4. Example C is caused by thiazide diuretics (hyperuricemia and hyperglycemia).
5. Example D is caused by beta-blockers. Look for a patient with a history of myocardial infarction (MI), heart failure, and/or hypertension who complains of increased fatigue and depression (avoid if possible in depressed patients, pulse <50 beats/min, second- or third-degree heart block).

U.S. CENTERS FOR DISEASE CONTROL AND PREVENTION STATISTICS**I. Discussion**

It is important that you memorize the following U.S. Centers for Disease Control and Prevention (CDC) statistics. The questions will be short and to the point. Determine whether a question is asking about the most common cause of death (mortality) or whether it is asking about the most common cause of a certain disease in a population (prevalence). For example, the most common cause of cancer death overall is lung cancer (mortality), but the most common cancer overall (prevalence) is skin cancer. The most common type of skin cancer is basal cell skin cancer.

Sometimes, a question will ask about a gender-specific cause. For example, the most common cancer in females is breast cancer, and the most common cancer in males is prostate cancer (prevalence). But the cancer causing the most deaths overall for both males and females is still lung cancer (mortality).

Mortality Statistics

- *Disease causing the most deaths overall:* Heart disease
- *Cancer with the highest mortality:* Lung cancer
- *Cancer with the highest mortality in males and females:* Lung cancer
- *Most common cause of death in adolescents:* Motor vehicle crashes

Prevalence

- *Most common cancer in females:* Breast cancer
- *Most common cancer in males:* Prostate cancer
- *Most common type of cancer overall (males/females):* Skin cancer
- *Most common type of skin cancer (males/females):* Basal cell cancer

- *Skin cancer with the highest mortality: Melanoma*
- *Gynecological cancer (vulva, vagina, cervix, uterus, ovary):*
 - Uterine/endometrial cancer (most common gynecological cancer)
 - Ovarian cancer (second most common gynecological cancer)

II. Example

What is the most common type of gynecological cancer?

- A. Uterine cancer
- B. Cervical cancer
- C. Breast cancer
- D. Ovarian cancer

III. Correct Answer: Option A

- A. Uterine cancer

IV. Question Dissection

Best Clues

- This question is based on your recall of facts that you memorized (rote memory).
- Rule out breast cancer because it is not considered a gynecological cancer.

NOTES

1. There may be a question about the gynecological cancers (see example). These types of cancers are located in the pelvis (labia, vagina, uterus, fallopian tubes, ovaries).
2. Breast cancer is not classified as a gynecological cancer.

BENIGN PHYSIOLOGIC VARIANTS

I. Discussion

A benign variant is a physiologic abnormality that does not interfere with bodily processes or functions. There are very few questions on benign variants. Some examples of the benign variants that have been seen on the exams include the geographic tongue, torus palatinus, and a split or fishtail uvula (see Chapter 5). Benign variants are listed under the appropriate organ system.

II. Example

A 45-year-old female patient complains of a sore throat. Upon examination, the NP notices a bony growth midline at the hard palate of the mouth. The patient denies any changes or pain. It is not red, tender, or swollen. She reports a history of the same growth for many years without any change. Which of the following conditions is most likely?

- A. Torus palatinus
- B. Geographic tongue
- C. Acute glossitis
- D. Leukoplakia

III. Correct Answer: Option A

- A. Torus palatinus

IV. Question Dissection

Best Clues

- The description of a chronic bony growth located midline in the hard palate indicates torus palatinus.
- Rule out glossitis, geographic tongue, and hairy leukoplakia because they are all located on the tongue and not on the hard palate (roof of the mouth).

NOTES

1. A torus palatinus is a benign growth of bone (an exostosis) located midline on the hard palate and covered with normal oral skin. It is painless and does not interfere with function.
2. A "geographic tongue" has multiple fissures and irregular smoother areas on its surface that make it look like a topographic map. The patient may complain of soreness on the tongue after eating or drinking acidic or hot foods.
3. Leukoplakia is not a benign variant. It appears as a slow-growing white plaque that has a firm to hard surface that is slightly raised on the tongue or inside the mouth. It is considered a precancerous lesion. It is due to chronic irritation of the skin or precancerous changes on the tongue and inside the cheeks. Its causes include poorly fitting dentures, chewing tobacco (snuff), and using other types of tobacco. Refer the patient for a biopsy because it can sometimes become malignant.
4. Oral hairy leukoplakia (OHL) of the tongue is a painless white patch (or patches) that appears corrugated. It is usually located on the lateral aspects of the tongue (or other areas inside the mouth) and is associated with HIV and AIDS infection. It is caused by Epstein-Barr virus (EBV) infection of the tongue. It is not considered a premalignant lesion.

U.S. PREVENTIVE SERVICES TASK FORCE SCREENING GUIDELINES

I. Discussion

U.S. Preventive Services Task Force (USPSTF) screening guidelines are graded as A, B, C, D, or I (insufficient evidence or evidence is lacking or of poor quality). The highest rating is a Grade A (routine screening is advised—high certainty that the net benefit is substantial). A rating of Grade D means that the harm outweighs the benefits (or there is no benefit) to the service and the use of the service is discouraged.

Both the ANCC and the AANPCB exams use the USPSTF screening recommendations. Regarding breast cancer screening, the USPSTF (2018) currently recommends that a screening baseline mammogram (with or without clinical breast exam) start at age 50 years and then every 2 years until the age of 74 years. For women aged 40 to 49 years, mammograms should be based on individual factors (such as risk factors, preferences, risk vs. benefits of mammograms).

II. Example

What is the USPSTF screening recommendation for ovarian cancer?

- A. Annual bimanual pelvic exam with pelvic ultrasound
- B. Pelvic and intravaginal ultrasound
- C. Intravaginal ultrasound with CA-125 tumor marker
- D. The USPSTF does not recommend routine screening of women for ovarian cancer (Grade D)

III. Correct Answer: Option D

D. The USPSTF does not recommend routine screening of women for ovarian cancer (Grade D)

IV. Question Dissection

Best Clues

- Do not "overread" the question. Assume that the question is asking about routine screening of the general population.
- Transvaginal ultrasound and CA-125 are not used for routine screening.
- Although the bimanual pelvic exam is "low tech," it is being used as a distractor.

NOTES

1. The USPSTF (2018) is against routine screening for ovarian cancer in asymptomatic women who are not known to have a high risk of hereditary cancer syndrome.
2. If there is a case scenario of an older woman who complains of vague abdominal/pelvic symptoms (stomach bloating, low-back ache, constipation) and is found to have a palpable ovary during the bimanual exam, rule out ovarian cancer.
3. Always rule out ovarian cancer in a postmenopausal woman who has a palpable ovary.
4. The next step is to order a pelvic and intravaginal ultrasound with CA-125. Refer to an oncologist.
5. The strongest risk factor for ovarian cancer (or breast cancer) is genetic predisposition, such as *BRCA1* or *BRCA2* mutations and positive family history (Table 2.3). Absolute risk of developing ovarian cancer is 35% to 45% in women with *BRCA1* mutation. Other risk factors are age, obesity, clomiphene (Clomid) use, and early-age menarche or late-onset menopause.

Table 2.3 USPSTF Screening Guidelines

Disease	Screening Test
Breast cancer (2018)	Baseline mammogram at age 50 years Screen every 2 years until age 74 years (biennial) After age 75 years (insufficient evidence)
Cervical cancer (2018)	Baseline Pap smear/cytology at age 21 years (do not screen younger) Screen every 3 years until age 65 years; at age 30–65 years, another option is to screen every 5 years using hrHPV testing alone or in combination with cytology (cotesting).
Lung cancer (2013)	Aged 55–80 years with 30 pack-year history of smoking or quit smoking up to 15 years previously Annual screening with LDCT
Hysterectomy (no cervix)	Do not screen (if no history of precancer or cervical cancer).
Prostate cancer (2018)	Aged 55–69 years, PSA-based screening should be individualized; ≥70 years, against prostate cancer screening.
Testicular cancer (2011)	Routine screening is not recommended.
Colorectal cancer (2016)	Baseline screening at age 50 years Use <i>high-sensitivity</i> fecal occult blood test (yearly) or sigmoidoscopy (every 5 years) or colonoscopy (every 10 years) from age 50 to 75 years; aged 76–85 years, individualize.
Skin cancer (2016)	Current evidence insufficient to assess benefits/harms of visual skin exam. Educate fair-skinned persons to avoid sunlight (10 a.m.–3 p.m.) and use sunblock ≥SPF 15.
Tobacco smoking (2020)	Ask all adults about tobacco use, advise them to stop smoking, and provide behavioral interventions/pharmacotherapy (if not pregnant) for smoking cessation.
Fall prevention in community-dwelling older adults (2018)	Adults aged ≥65 years, exercise interventions to prevent falls in adults at increased risk; against vitamin D supplementation.
Ovarian cancer (2018)	Routine screening is not recommended. High-risk (<i>BRCA</i> mutation, family history of breast/ovarian cancer) are screened by specialist; refer for genetic counseling.
Abdominal aortic aneurysm (2019)	One-time screening (men aged 65–75 years) for cigarette smokers or those who have quit. Screening test is ultrasound of abdomen.

hrHPV, high-risk human papillomavirus; LDCT, low-dose computed tomography; SPF, sun protection factor; USPSTF, U.S. Preventive Services Task Force.

FOLLOWING UP ON A PRESCRIPTION MEDICINE

I. Discussion

In these cases, a patient who is taking prescription medicine is running out of their supply or does not have refills left. The test-taker must decide what type of initial follow-up is needed. Depending on the case scenario, for a patient who is fully symptomatic due to the abrupt cessation of medicine (either due to running out of refills or due to discontinuation of health insurance), a reasonable initial action is to continue the prescription medicine.

II. Example

A 65-year-old female smoker presents with a history of Barrett's esophagus and gastroesophageal reflux disease (GERD). The patient reports that her gastroenterologist's prescription for esomeprazole (Nexium) 40 mg daily ran out a few days ago. She is complaining of severe heartburn and a sore throat. During the physical exam, the NP notes an erythematous posterior pharynx without tonsillar discharge and mild dental enamel loss on the rear molars. What is the best initial action for the NP to follow?

- A. Refer the patient to an oncologist for a biopsy to rule out esophageal cancer.
- B. Give the patient a refill of her proton-pump inhibitor (PPI) prescription and advise her to schedule an appointment with her gastroenterologist.
- C. Recommend that the patient take over-the-counter (OTC) ranitidine (Zantac) twice a day until she can be seen by her gastroenterologist.
- D. Switch the patient's prescription to another brand of PPI because her symptoms are not improving.

III. Correct Answer: Option B

- B. Give the patient a refill of her proton-pump inhibitor (PPI) prescription and advise her to schedule an appointment with her gastroenterologist.

IV. Question Dissection

Best Clues

- Rule out option A because the patient is already under the care of a gastroenterologist.
- OTC ranitidine (Zantac) is not potent enough to control the symptoms of erosive esophagitis. A PPI is the preferred treatment for erosive esophagitis.
- Do not switch the patient to another brand of PPI. Her worsening symptoms are due to rebound caused by abrupt cessation of PPI.
- The best initial action in this case is to refill the PPI prescription because the patient is fully symptomatic (erosive esophagitis) until she can see her gastroenterologist.

NOTES

1. The patient's severe symptoms are caused by the sudden discontinuation of the high-dose PPI (rebound type of reaction).
2. Barrett's esophagus is the "precancerous" lesion of esophageal cancer. It is best managed by a gastroenterologist (not an oncologist).
3. Patients diagnosed with Barrett's esophagus typically have endoscopic examinations with biopsy by a gastroenterologist annually (or every 6 months for high-grade lesions).
4. Patients with Barrett's esophagus are treated with high-dose PPIs for a "lifetime."
5. The first-line treatment of mild, uncomplicated GERD is lifestyle changes (e.g., avoid eating 3–4 hours before bedtime, dietary changes, weight loss if overweight).
6. If a patient is at high risk for esophageal cancer (aged 50 years or older, smoker, chronic GERD for decades), consider referral to a gastroenterologist for an upper endoscopy.

TANNER STAGES

I. Discussion

Expect to see at least one or two questions regarding Tanner stages in girls and boys. Because Tanner stage I is prepuberty and Tanner stage V is the adult pattern for both boys and girls, the only stages to memorize for the exams are Tanner stage II to Tanner stage IV.

For girls, memorize the pattern of breast development, and for boys, the genital development (testes and penis). It is not as important to memorize pubic hair development for both.

Tanner Stages

Girls

- *Stage I*: Prepubertal pattern
- *Stage II*: Breast bud and areola start to develop.
- *Stage III*: Breast continues to grow with nipples/areola (one mound/no separation).
- *Stage IV*: Nipples and areola become elevated from the breast (a secondary mound).
- *Stage V*: Adult pattern

Boys

- *Stage I*: Prepubertal pattern
- *Stage II*: Testes and scrotum start to enlarge (scrotal skin starts to get darker/more rugae).
- *Stage III*: Penis grows longer (length) and testes/scrotum continues to become larger.
- *Stage IV*: Penis become wider and continues growing in length (testes are larger with darker scrotal skin and more rugae).
- *Stage V*: Adult pattern

II. Example

A 14-year-old boy is brought in by his mother for a physical exam. Both are concerned about his breast enlargement. The teen denies breast tenderness. On physical exam, the NP palpates soft breast tissue that is not tender. No dominant mass is noted. The skin is smooth, and there is no nipple discharge with massage. The teen has a body mass index (BMI) of 29. Which of the following statements is correct?

- A. Advise the mother that the patient has physiologic gynecomastia and should return for a follow-up exam.
- B. Order an ultrasound of both breasts to further assess the patient's breast tissue development.
- C. Reassure the mother that the patient's breast development is within normal limits.
- D. Educate the mother that her son has pseudogynecomastia.

III. Correct Answer: Option D

- D. Educate the mother that her son has pseudogynecomastia.

IV. Question Dissection

Best Clues

- The boy is very overweight (BMI 29) and is almost obese.
- The clinical breast exam does not show palpable breast tissue. Instead, the breast palpation reveals soft fatty tissue.
- It is wrong to "reassure" a patient or family member in the exam (poor therapeutic communication technique).

NOTES

1. Physiologic gynecomastia physical exam findings will show disklike breast tissue that is mobile under each nipple/areola; the breast may be tender, and the breast can be asymmetrical (one breast larger than the other).
2. A BMI of 25 to 29.9 is considered overweight. Obesity is a BMI of 30 or higher.
3. Overweight to obese males are at highest risk for pseudogynecomastia.

QUESTIONS ABOUT FOOD

I. Discussion

There are basically three kinds of food-related questions on the exam. You may be asked to pick the foods that have high levels of certain minerals, such as potassium, calcium, or magnesium. Other questions

will address food interactions (tetracycline and dairy), drug interactions (monoamine oxidase inhibitor [MAOI] and high-tyramine-containing foods such as fermented foods), or foods that should be avoided for a particular disease (e.g., wheat products in the case of celiac disease).

Certain foods are recommended for certain diseases (e.g., salmon/omega-3 for heart disease) because of their favorable effect. In contrast, certain foods are contraindicated for some conditions because of their adverse or dangerous effects.

II. Example

Which of the following foods are known to have high potassium content?

- A. Low-fat yogurt, soft cheeses, and collard greens
- B. Aged cheese, red wine, and chocolate
- C. Potatoes, apricots, and Brussels sprouts
- D. Black beans, red meat, and citrus juice

III. Correct Answer: Option C

C. Potatoes, apricots, and Brussels sprouts

IV. Question Dissection

Best Clues

- First, look at the answer options for inconsistencies in the list of foods.
- Rule out option A because it is inconsistent and these foods do not contain high levels of potassium: low-fat yogurt and soft cheeses (calcium) with collard greens (vitamin K).
- Rule out option B because these foods have a high tyramine, not potassium, content.
- Rule out option D because it is inconsistent. Although citrus juices are high in potassium, both black beans and red meat are not (iron).
- If options A, B, and D are incorrect, then the only one left is option C (potatoes, apricots, and Brussels sprouts). A large number of fruits and vegetables are rich in potassium and vitamins.

NOTES

1. Foods with a high tyramine content can cause dangerous food–drug interactions with MAOI inhibitors (e.g., isocarboxazid [Marplan], phenelzine [Nardil], and tranylcypromine [Parnate]).
2. Foods and supplements containing stimulants, such as caffeine and ephedra, are best avoided by patients with hypertension, arrhythmias, high risk for MI, hyperthyroid disease, albuterol use, amphetamine use, and so on.
3. If one of the food choices in an answer option is incorrect, rule out this option because all of the foods on the list have to be correlated.

Examples of Food Groups

1. *Gluten (avoid with celiac disease/celiac sprue)*: Wheat (including spelt and kamut), rye, barley (breads, cereals, pasta, cookies, cakes)
 - *Gluten free (safe carbohydrates)*: Corn, rice, potatoes, quinoa, tapioca, soybeans
2. *Plant sterols and stanols (reduce cholesterol, LDL, triglycerides)*: Sterol-fortified spreads (Benecol spread), sterol-fortified foods, wheat germ, sesame oil
3. *Monounsaturated fats/fatty acids (decrease risk of heart disease)*:
 - Olive oil, canola oil, some nuts (almonds, walnuts), sunflower oil/seeds
 - Mediterranean diet, which is high in monounsaturated fats
4. *Saturated fats or trans fats (increase risk of heart disease)*: Lard, beef fat (fatty steak), deep-fried fast foods
5. *Omega-3 or fish oils (decrease risk of heart disease)*: Fatty cold-water marine fish (salmon), fish oils, flax-seed oil, and krill oil
6. *Magnesium (decreases BP, dilates blood vessels)*: Some nuts (almonds, peanuts, cashews), some beans, whole wheat; also found in laxatives, antacids, milk of magnesia

7. *Potassium (helps decrease BP)*: Most fruits (especially apricot, banana, orange, prune juice), some vegetables
8. *Folate (decreases homocysteine levels and fetal neural tube defects)*: Breakfast cereals fortified with folate, green leafy vegetables (i.e., spinach), liver
9. *Iron (treats iron-deficiency anemia)*: Beef, liver, black beans, black-eyed peas
10. *Vitamin K (should control intake if on anticoagulants)*: Green leafy vegetables (kale, collard greens, spinach), broccoli, cabbage
11. *High sodium content (increases water retention, can increase BP)*: Cold cuts, pickles, preserved foods, canned foods, hot dogs, chips
12. *Calcium (helps with osteopenia and osteoporosis, helps decrease BP)*: Low-fat dairy, low-fat milk, low-fat yogurt, cheeses

Common Disorders Associated With Certain Foods

1. Celiac disease
 - Lifetime avoidance of gluten-containing cereals such as wheat, rye, and barley is necessary.
 - *Gluten free*: Rice, corn, potatoes, peanuts, soybeans, meat, dairy, all fruits/vegetables; most people with celiac disease can eat oats.
2. Hypertension
 - Maintain an adequate intake of calcium, magnesium, and potassium.
 - *Calcium*: Low-fat dairy, low-fat yogurt, cheeses
 - *Magnesium*: Wheat bread, nuts (almonds, peanuts, cashews), some beans
 - *Potassium*: Most fruits (apricot, banana, oranges, cantaloupes, raisins), green vegetables
 - *Avoid high-sodium foods*: Cold cuts, pickles, preserved foods, canned foods, preservatives
3. Migraine headaches and MAOIs (Marplan, Nardil, and Parnate)
 - *High-tyramine foods*: Aged cheeses/meats, red wine, fava beans, draft beer, fermented foods
4. Anticoagulation therapy (i.e., warfarin sodium or Coumadin)
 - Avoid eating large amounts of leafy green vegetables (kale/collard greens, spinach, cabbage, broccoli) and cooking with canola oil (high in vitamin K); use other oils instead. High levels of vitamin K decrease the effects of warfarin sodium.

CASE MANAGEMENT

I. Discussion

Medical case managers are experienced RNs who work for hospitals, healthcare plans, and health insurance companies. Their job is essentially to coordinate the outpatient healthcare of patients with high-cost chronic conditions. The goal of case management is to decrease disease exacerbations and decrease the risk of hospitalization.

Asthma

A good outcome for children with asthma is their ability to attend school full time and to play normally every day. A poor case management outcome is if the child misses school and/or is unable to play because of poor control of asthma symptoms.

NOTES

1. With asthma, a good case outcome is for the child (or adult) to return to normal function. For children, it means the child is attending school and can play daily. If the child is not able to attend school full time (frequent absences), this is a poor case management outcome.
2. The risk factors for asthma fatality include a history of ED visits, frequent use of short-acting beta-agonist use (i.e., albuterol), nocturnal awakenings, increased dyspnea and wheezing, respiratory viral infection, and so on.
3. Diseases that are selected for case management are usually chronic conditions, such as asthma, congestive heart failure (CHF), HIV infection, and chronic psychiatric conditions. A good outcome will show good symptom control and no exacerbations or hospitalizations.

ALL QUESTIONS HAVE ENOUGH INFORMATION

I. Discussion

Assume that all the questions on the exams contain enough information to answer them correctly. Do not read too much into a question or assume that it is missing some vital information. As far as the ANCC and AANPCB exams are concerned, all questions contain enough information to allow you to solve them correctly. Consider a patient is in good health unless a disease or other health condition is mentioned in the test question.

EMERGENT CASES

I. Discussion

The ability to recognize and initially manage emergent conditions that may present in the primary care arena is a skill that is expected of all NPs. It is important to memorize not only the presenting signs and symptoms of a given condition but also its initial management in primary care.

Learn how these conditions present so that you can recognize them in the exam. The following is a list of emergent conditions that will be on the exam. (They are discussed in detail in the “Danger Signals” sections of Chapters 5 through 17.)

Danger Signals

Cardiovascular System

- Acute MI, unstable angina
- Heart failure
- Bacterial endocarditis
- Abdominal aortic aneurysm (AAA)

Skin and Integumentary System

- Angioedema/anaphylaxis
- Stevens–Johnson syndrome
- Meningococemia
- Rocky Mountain spotted fever (RMSF)
- Lyme disease
- Herpes zoster ophthalmicus
- Melanoma
- Basal cell carcinoma

Gastrointestinal System

- Acute abdomen (surgical abdomen)
- Acute appendicitis
- Acute pancreatitis
- Acute diverticulitis
- Acute cholecystitis
- *Clostridium difficile* infection
- Colon cancer

Men's Health

- Testicular torsion
- Priapism
- Prostate cancer

Psychosocial Mental Health

- Depression with suicidal plan
- Acute serotonin syndrome
- Malignant neuroleptic syndrome

Nervous System

- Cerebrovascular accident (CVA)
- Temporal arteritis headache
- Subarachnoid bleeding
- Acute bacterial meningitis
- Subdural hematoma

Head, Eyes, Ears, Nose, and Throat

- Retinal detachment
- Orbital cellulitis
- Optic neuritis
- Peritonsillar abscess
- Battle sign
- Herpes keratitis
- Temporal arteritis
- Acute angle-closure glaucoma

Pulmonary System

- Anaphylaxis
- Severe asthmatic exacerbation (impending respiratory failure)
- Pulmonary emboli
- CAP
- Lung cancer

Renal System

- Acute pyelonephritis
- Acute renal failure
- Bladder cancer

Women's Health

- Dominant breast mass attached to surrounding tissue
- Ruptured tubal ectopic pregnancy
- Paget's disease of the breast
- Ovarian cancer

II. Example

An asthmatic male patient complains of a sudden onset of itching and coughing after taking two aspirin tablets for a headache in the waiting room. The patient's lips and eyelids are becoming swollen. The patient complains of feeling hot. Bright-red wheals are noted on his chest and arms and legs. Which of the following is the best initial intervention to follow?

- A. Call 911.
- B. Check the patient's BP, pulse, and temperature.
- C. Give an injection of aqueous epinephrine 1:1,000 (1 mg/mL) 0.5 mg IM into the vastus lateralis muscle immediately.
- D. Initiate a prescription of a potent topical steroid and a Medrol Dose Pack.

III. Correct Answer: Option C

- C. Give an injection of aqueous epinephrine 1:1,000 dilution (1 mg/mL) 0.5 mg IM into the vastus lateralis muscle immediately.

IV. Question Dissection***Best Clues***

- The quick onset of symptoms, such as angioedema, after taking aspirin is a clue.
- The classic signs and symptoms of anaphylaxis described in this case should be noted.
- Severe anaphylactic episodes occur almost immediately or within 1 hour after exposure.

NOTES

1. *Treatment of anaphylaxis (in primary care):*
 - *If only one clinician is present:* Give an injection of epinephrine 1:1,000 dilution 0.3 to 0.5 mg intramuscularly *stat*, and then call 911. May repeat dose within 5 minutes in case of poor response.
2. *ED treatment medications:* Administer epinephrine IM, 100% oxygen by face mask, an antihistamine (H_1 antagonist) such as diphenhydramine (Benadryl), an H_2 antagonist such as ranitidine, a bronchodilator such as albuterol (short-acting beta-2 agonist), and systemic glucocorticosteroids such as prednisone.
3. Patients with an atopic history (asthma, eczema, allergic rhinitis) with nasal polyps are at higher risk for aspirin and nonsteroidal anti-inflammatory drug (NSAID) allergies.
4. Anaphylaxis is classified as a type I IgE-dependent reaction.
5. Biphasic anaphylaxis occurs in up to 23% of cases (symptoms recur within 8–10 hours after initial episode). This is the reason why these patients are prescribed a Medrol Dose Pack and a long-acting antihistamine after being discharged from the ED.
6. The most common triggers for anaphylaxis in children are foods. Medications and insect stings are the most common triggers in adults.

PRIORITIZING OTHER EMERGENT CASES

I. Discussion

During life-threatening situations, managing the airway, breathing, and circulation (the ABCs) is always the top priority. If the question does not describe conditions requiring the ABCs, then the next level of priority is the acute or sudden change in the mental status and the level of consciousness (LOC). One of the most important clues in such problems is the acute timing of onset of symptoms or the sudden change of the LOC from the patient's "baseline."

II. Example

A 16-year-old boy presents to a community clinic accompanied by his grandmother, who reports that he fell off his bike this morning. The patient now complains of a headache with mild nausea. The patient's grandmother reports that he was not wearing a helmet. The health history is uneventful. Which of the following statements is indicative of an emergent condition?

- A. The patient complains of multiple painful abrasions that are bleeding on his arms and legs.
- B. The patient complains of a headache that is relieved by acetaminophen (Tylenol).
- C. The patient makes eye contact occasionally and answers with brief statements.
- D. The patient is having difficulty with following normal conversation and answering questions.

III. Correct Answer: Option D

- D. The patient is having difficulty with following normal conversation and answering questions.

IV. Question Dissection

Best Clues

- History of recent trauma that is followed by a headache with nausea.
- The patient did not wear a bicycle helmet.

NOTES

1. Any recent changes in LOC, even one as subtle as difficulty with normal conversation, should ring a bell in your head.
2. Notice the words "normal conversation." Do not overread the question and ask yourself what they mean by "normal conversation." Take it at face value.
3. Changes in LOC on the test are usually subtle changes. Signs to watch for include difficulty answering questions, slurred speech, apparent confusion, inability to understand instructions/conversation, being sleepy/lethargic, and so forth.
4. Even though the patient is bleeding, note that he has "abrasions," which are superficial.
5. The behavior described in option C is considered "normal" for an adolescent male (or female).